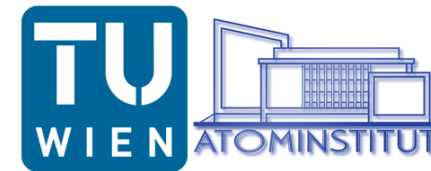


Towards Detecting Quantum Field Entanglement at Finite Temperatures

Taufiq Murtadho



Theory: Marek Gluza, Nelly Ng

Experiment: Jörg Schmiedmayer, Sebastian Erne, Federica Cataldini, Mohammadamin Tajik

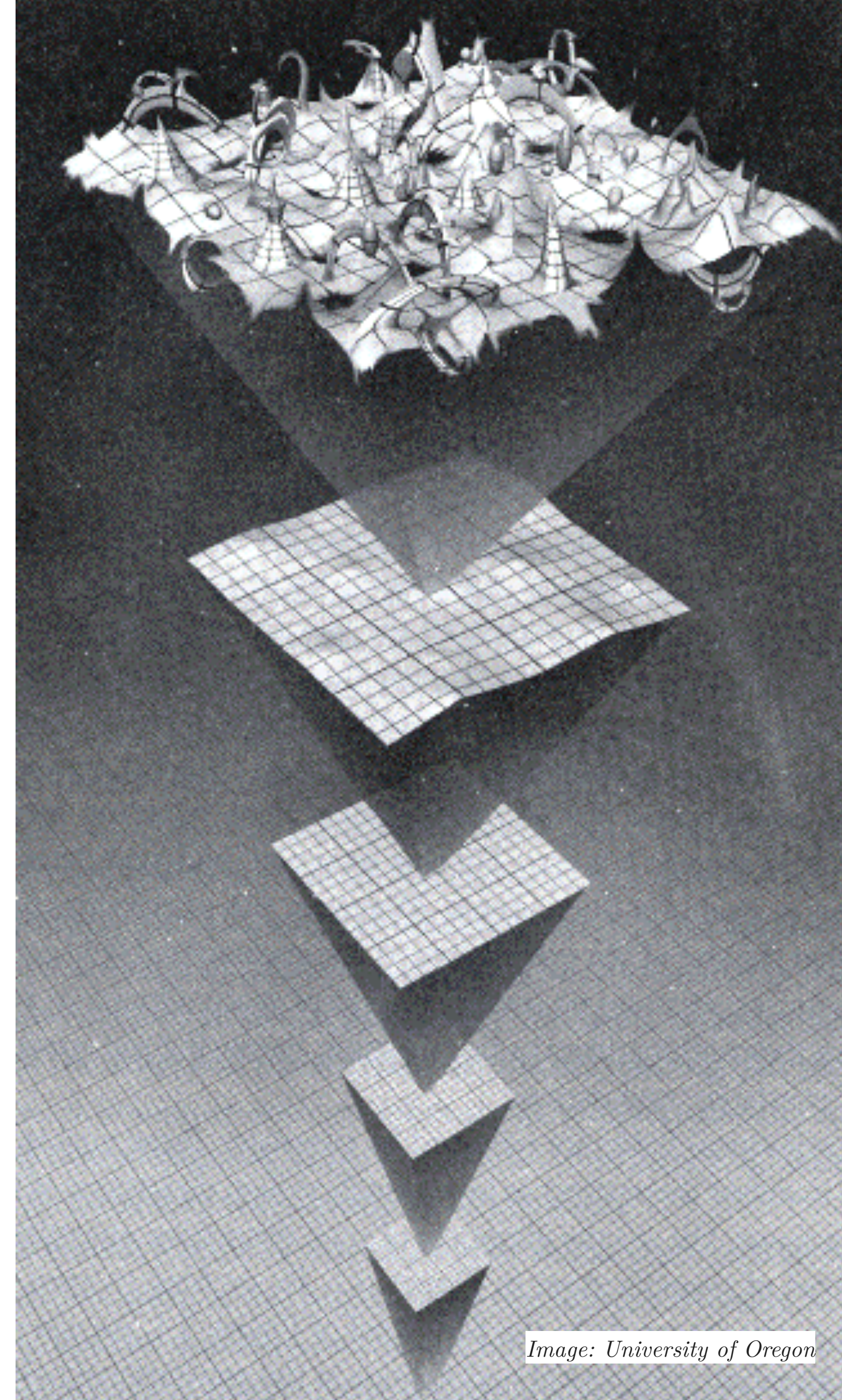


Image: University of Oregon

Measurement of total phase fluctuation in cold-atomic quantum simulators

Taufiq Murtadho ^{1,*} Federica Cataldini ² Sebastian Erne ² Marek Gluza ¹ Mohammadamin Tajik ²
Jörg Schmiedmayer ² and Nelly H. Y. Ng ^{1,†}

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IOP Publishing

Quantum Sci. Technol. 11 (2026) 035060

<https://doi.org/10.1088/2058-9565/ae8b1d>

Quantum Science and Technology

 Check for updates

PAPER

Extensive entanglement between coupled Tomonaga–Luttinger liquids in and out of equilibrium

Taufiq Murtadho ¹, Marek Gluza ^{1,*} and Nelly H Y Ng ^{1,2,*}

¹ School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore 637371, Singapore

² Centre for Quantum Technologies, Nanyang Technological University Singapore, 50 Nanyang Avenue, Singapore 639798, Singapore

* Authors to whom any correspondence should be addressed.

E-mail: fiqmurtadho@gmail.com and nelly.ng@ntu.edu.sg

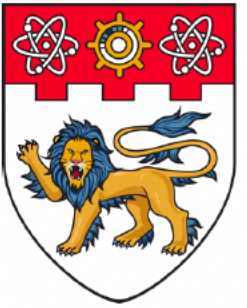
Keywords: quantum entanglement, quantum gases, Tomonaga–Luttinger liquids

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NANYANG
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SINGAPORE

RECONSTRUCTING HIDDEN
QUANTUM FIELDS:
MEASUREMENT AND
TOMOGRAPHY IN ULTRACOLD
BOSE GASES

MUHAMMAD TAUFIQ MURTADHO

→ Preliminary Direction, Questions
& Invitation to Collaborate

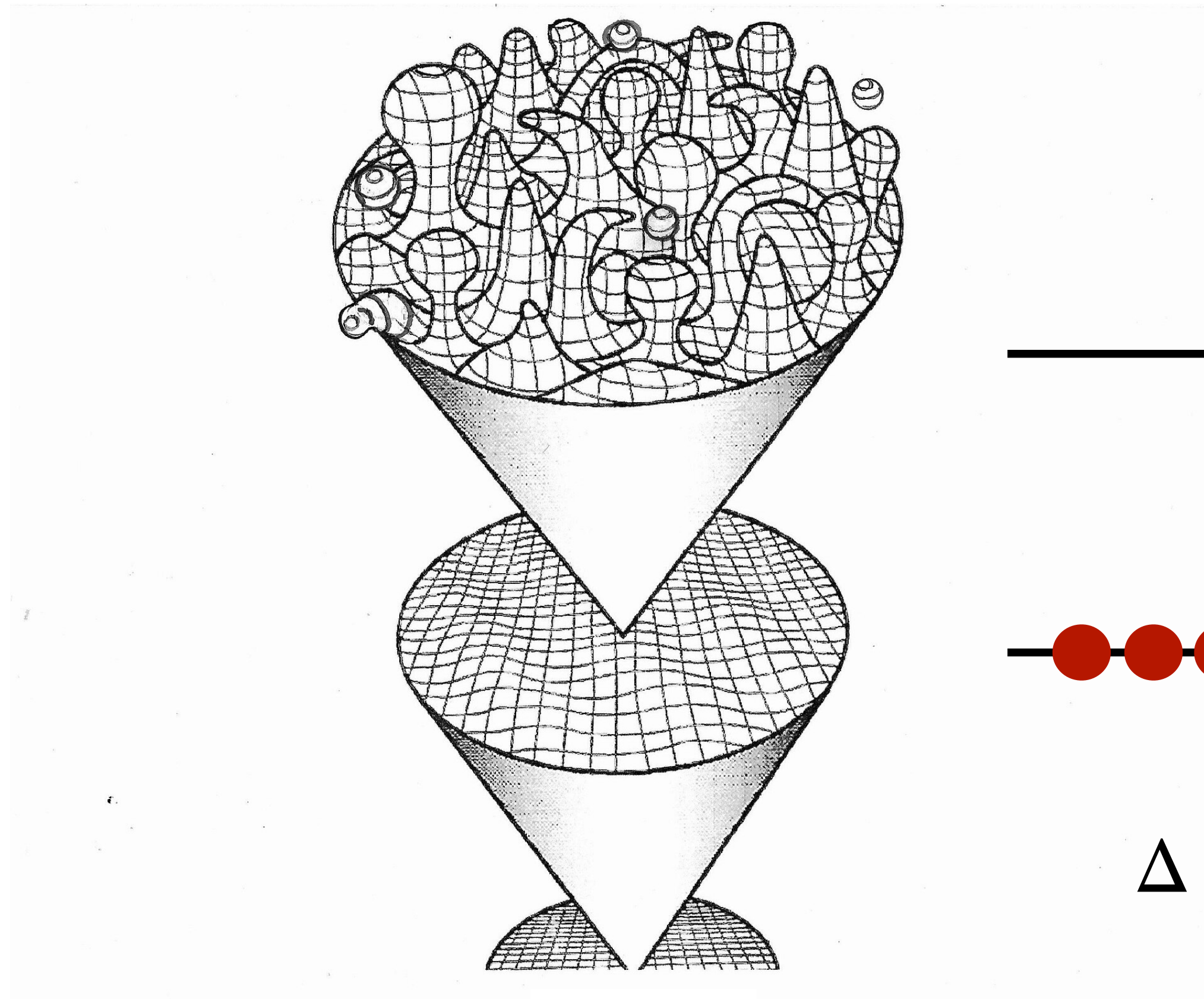
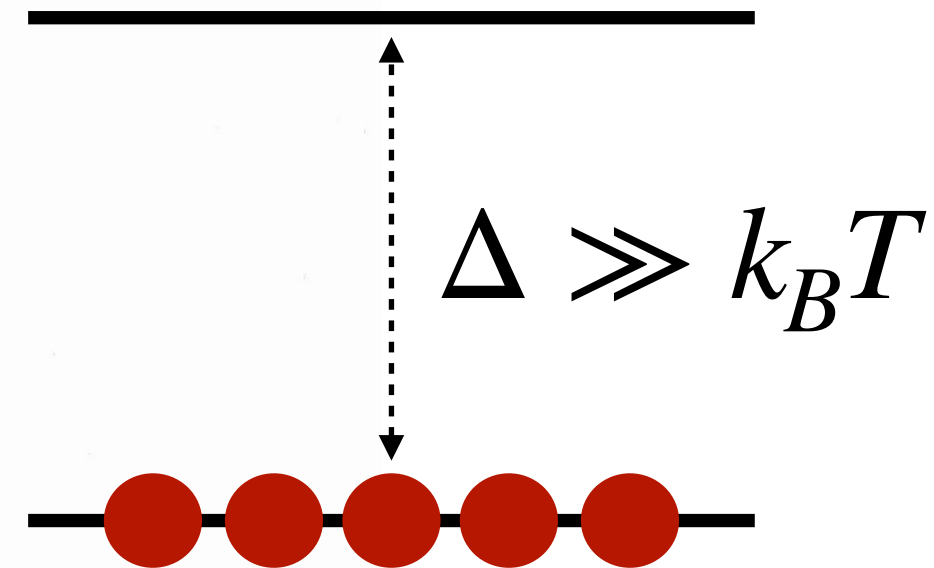


Image: Wikimedia



$$\Delta \rightarrow 0?$$

We do **not** live in a quantum vacuum

Quantum effects live in thermal background

Fluctuation in Bose-Einstein Condensate (BEC)

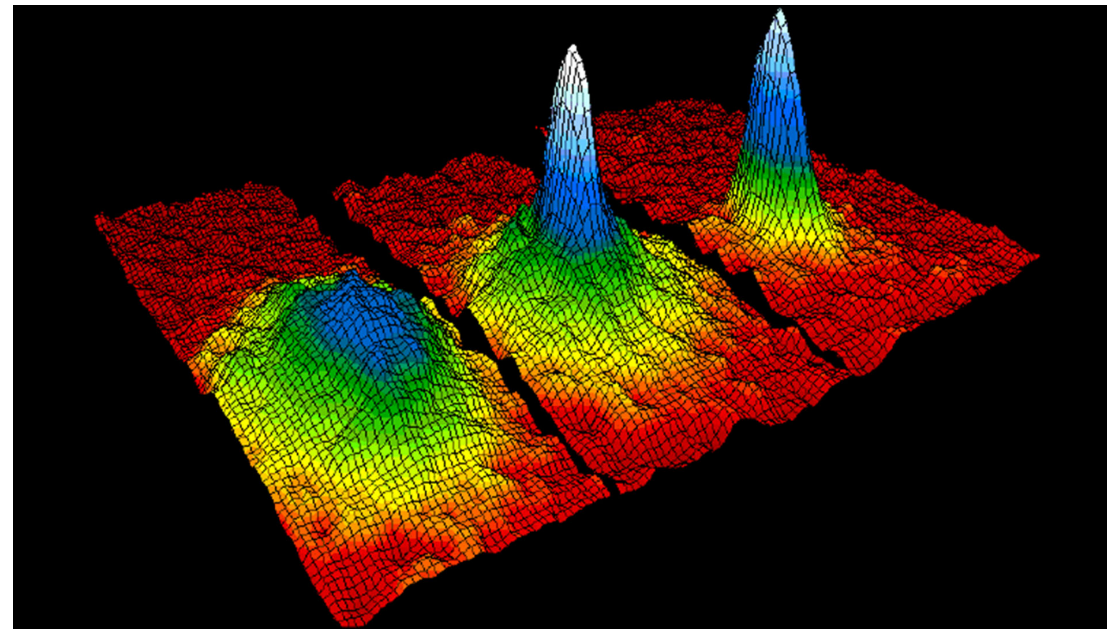
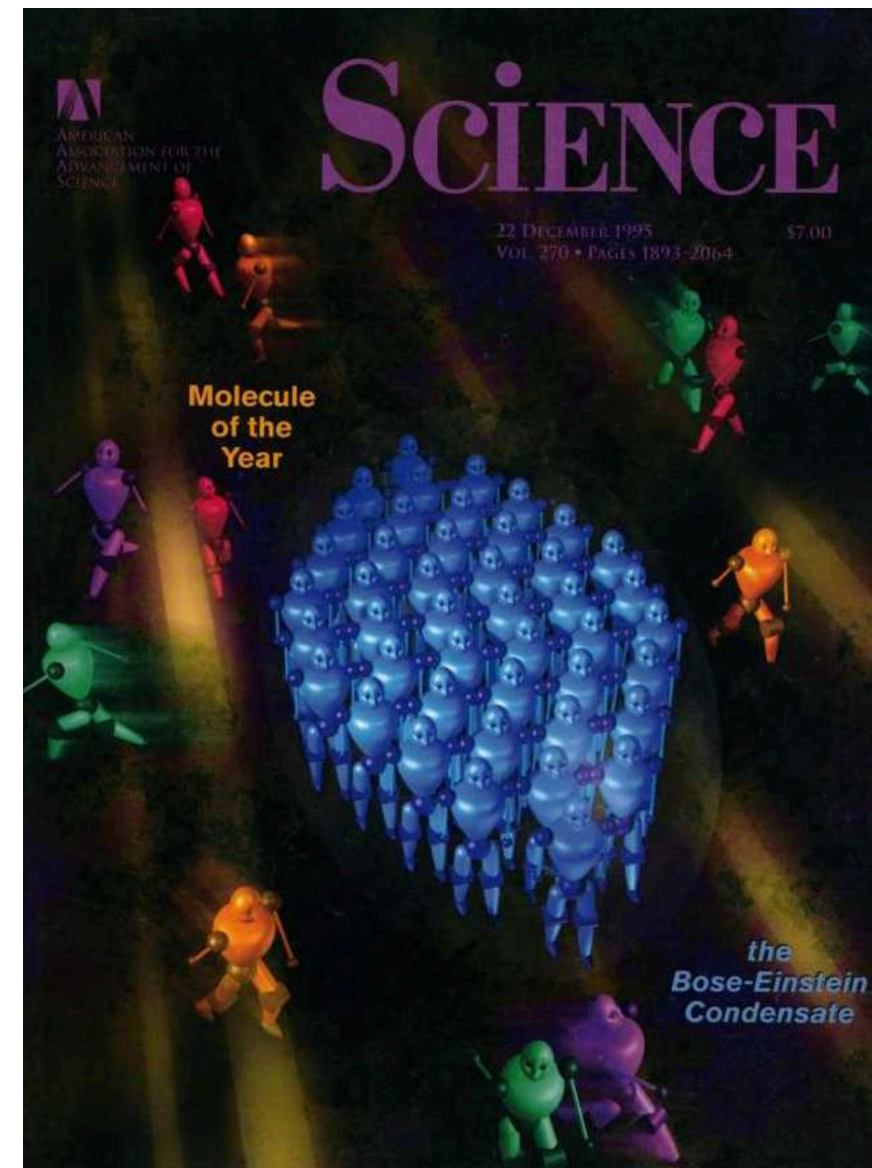


Image: NIST

Anderson, M. H., Ensher, J. R., Matthews, M. R., Wieman, C. E., & Cornell, E. A. *Science*, 269(5221), 198-201 (1995).

Davis, K. B., Mewes, M. O., Andrews, M. R., van Druten, N. J., Durfee, D. S., Kurn, D. M., & Ketterle, W. *PRL*, 75(22), 3969 (1995).

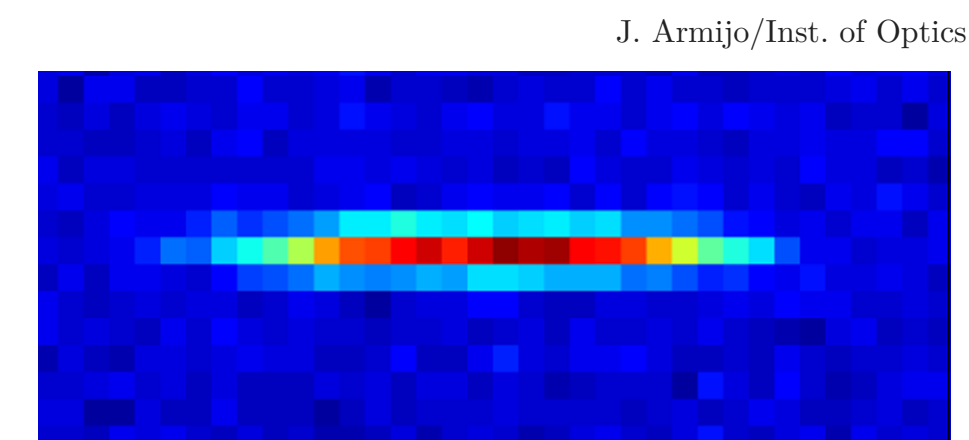


Science Magazine, 22 Dec. 1995

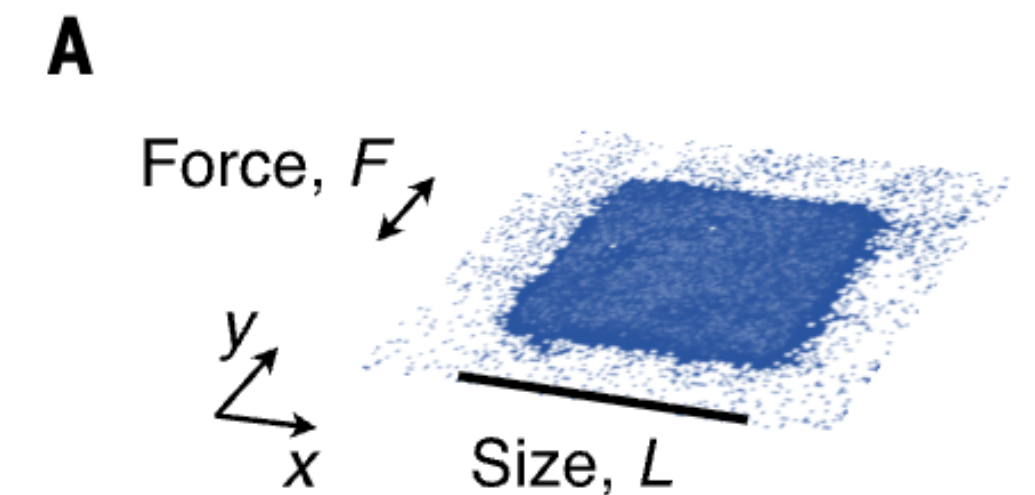
$$\varepsilon_k \approx kc$$

$$\varepsilon_k \rightarrow 0 \text{ as } k \rightarrow 0 \text{ (Goldstone)}$$

Fluctuation is enhanced in low-dimension (1D/2D)



Armijo, J. *PRL*, 108(22), 225306 (2012).



Gazo, M., Karailiev, A., Satoor, T., Eigen, C., Galka, M., & Hadzibabic, Z. *Science*, 389(6762), 802-805 (2025).

$$\int \frac{d^d k}{k^2} \sim \begin{cases} k^{-1} & d = 1 \\ \log k & d = 2 \\ \text{finite} & d = 3 \end{cases}$$

No long-range order at any finite temperature in thermodynamic limit for $d = 1$ and $d = 2$

Finite Temperature Quantum Field Theory (QFT)

Rapid development in the past few decades:

Le Bellac, M. Thermal field theory. Cambridge University Press (2000).

Kraemmer, U., & Rebhan, A. Advances in perturbative thermal field theory. *Reports on Progress in Physics*, 67(3), 351-431 (2004).

Laine, M., & Vuorinen, A. Basics of thermal field theory. *Lect. Notes Phys*, 925(1), 1701-01554 (2016).

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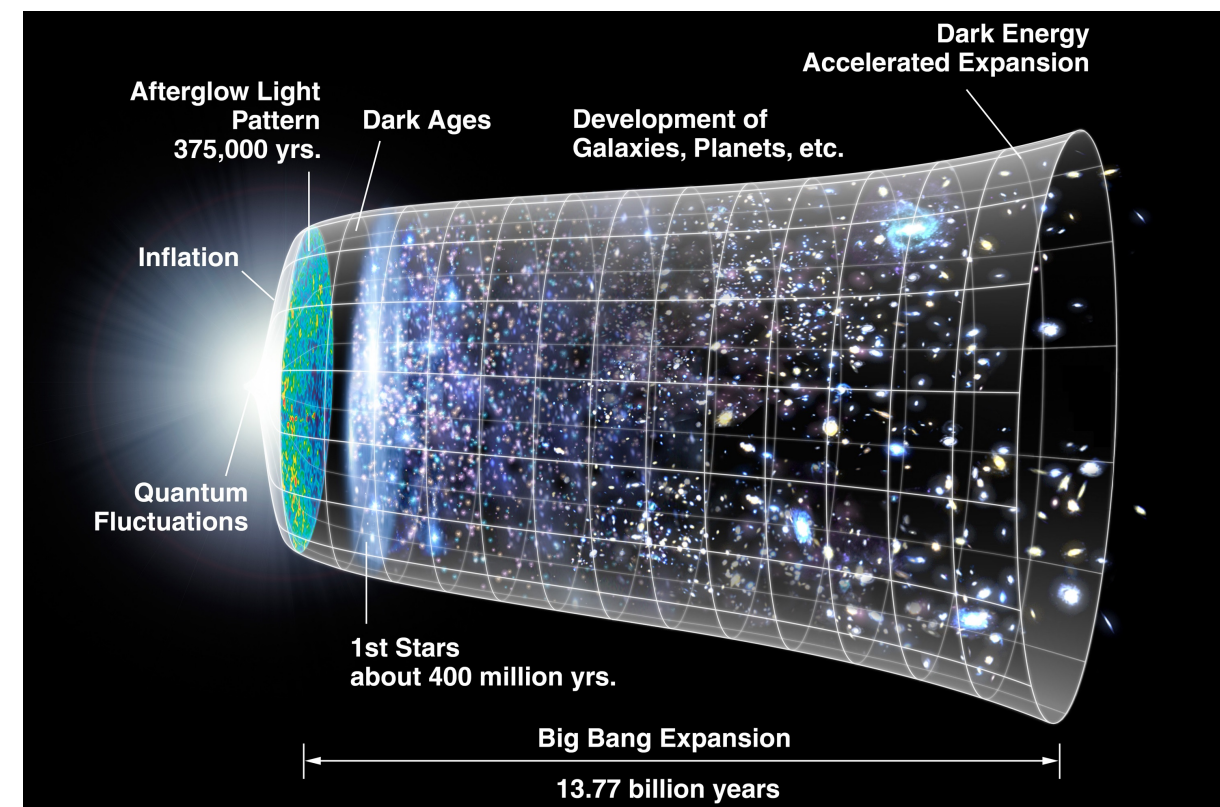


Image: Wikimedia

Big Bang & Early Universe

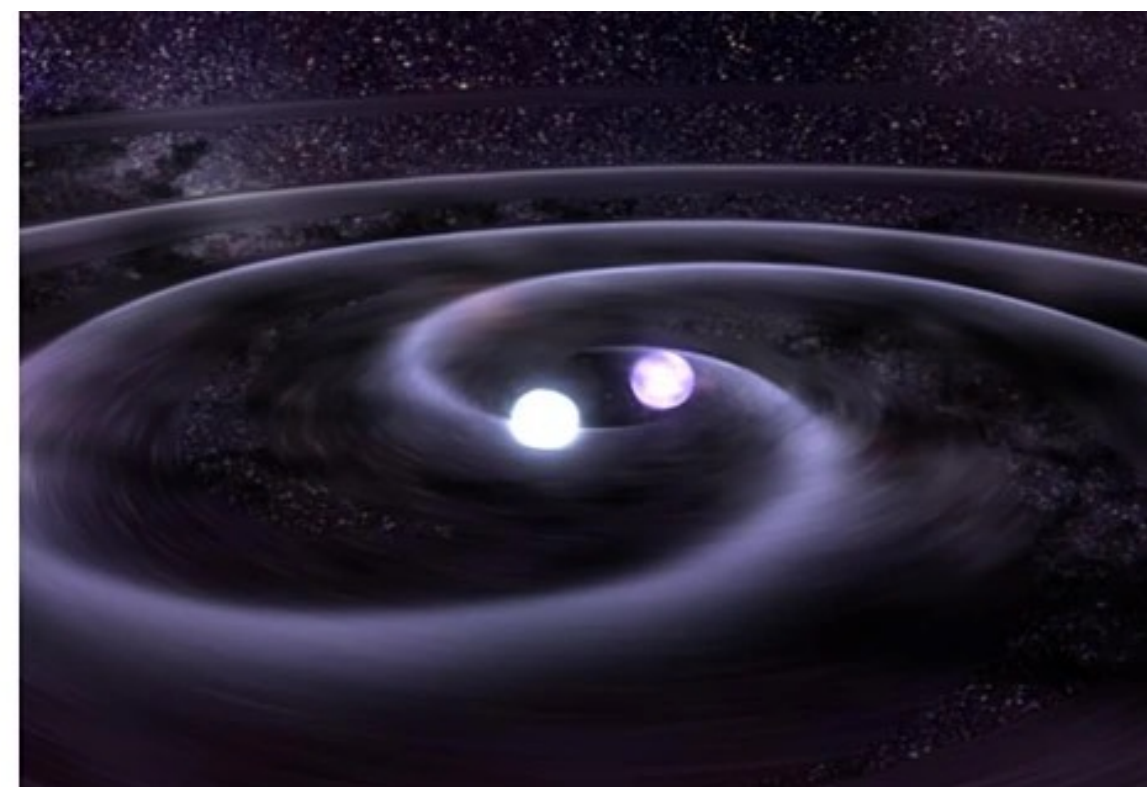


Image: Princeton

Neutron stars merger

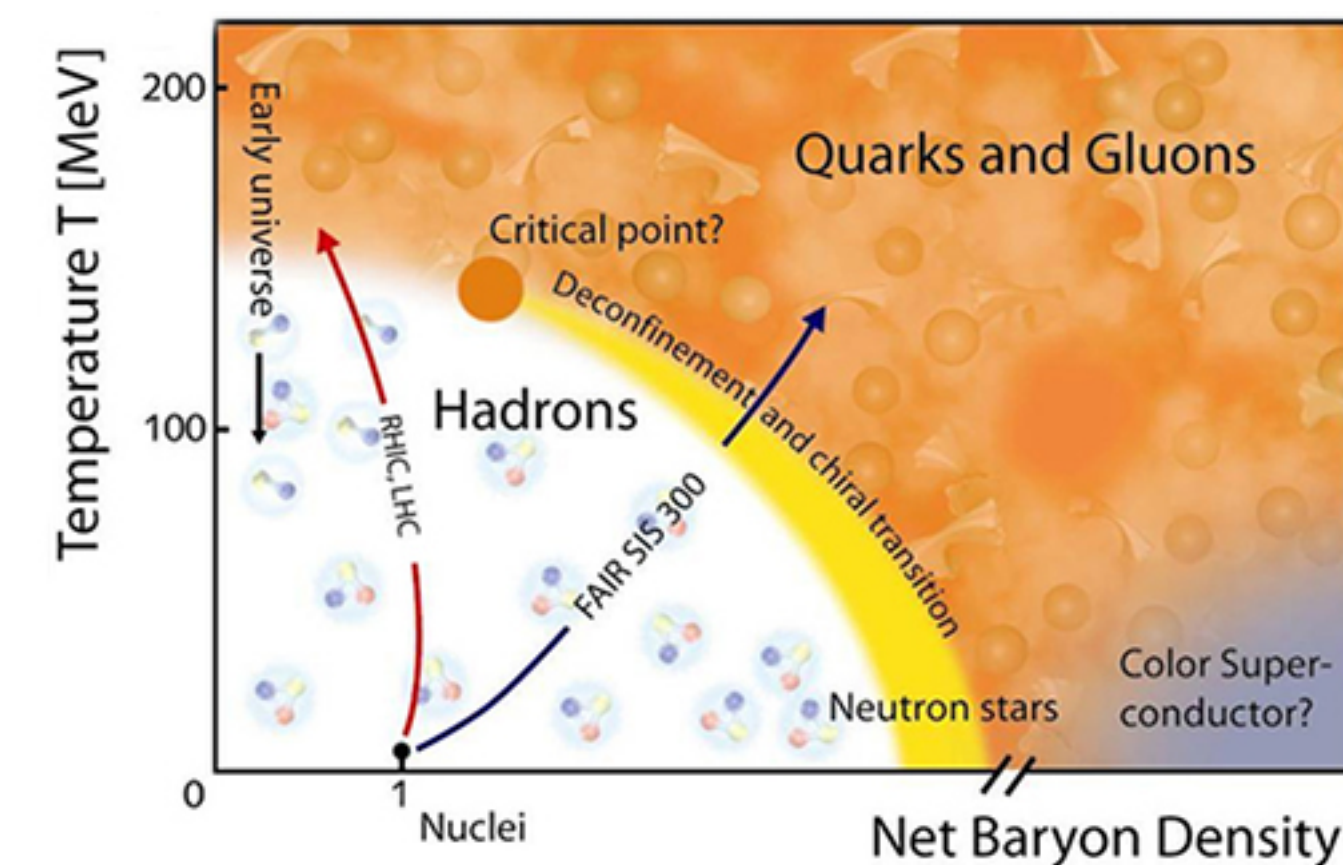
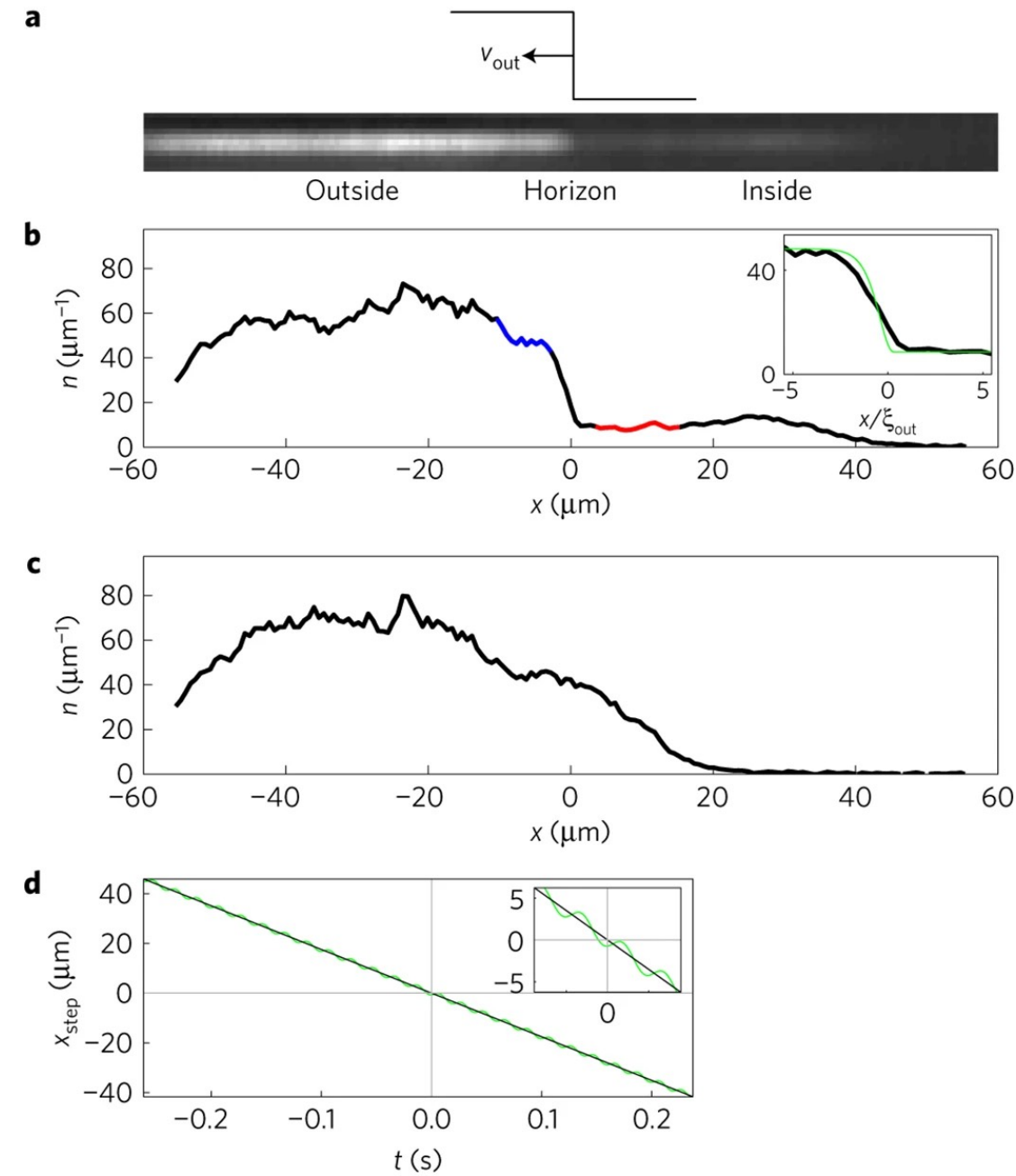


Image: CERN

Quark-Gluon Plasma

Quantum Field Simulation with Bosonic Gases

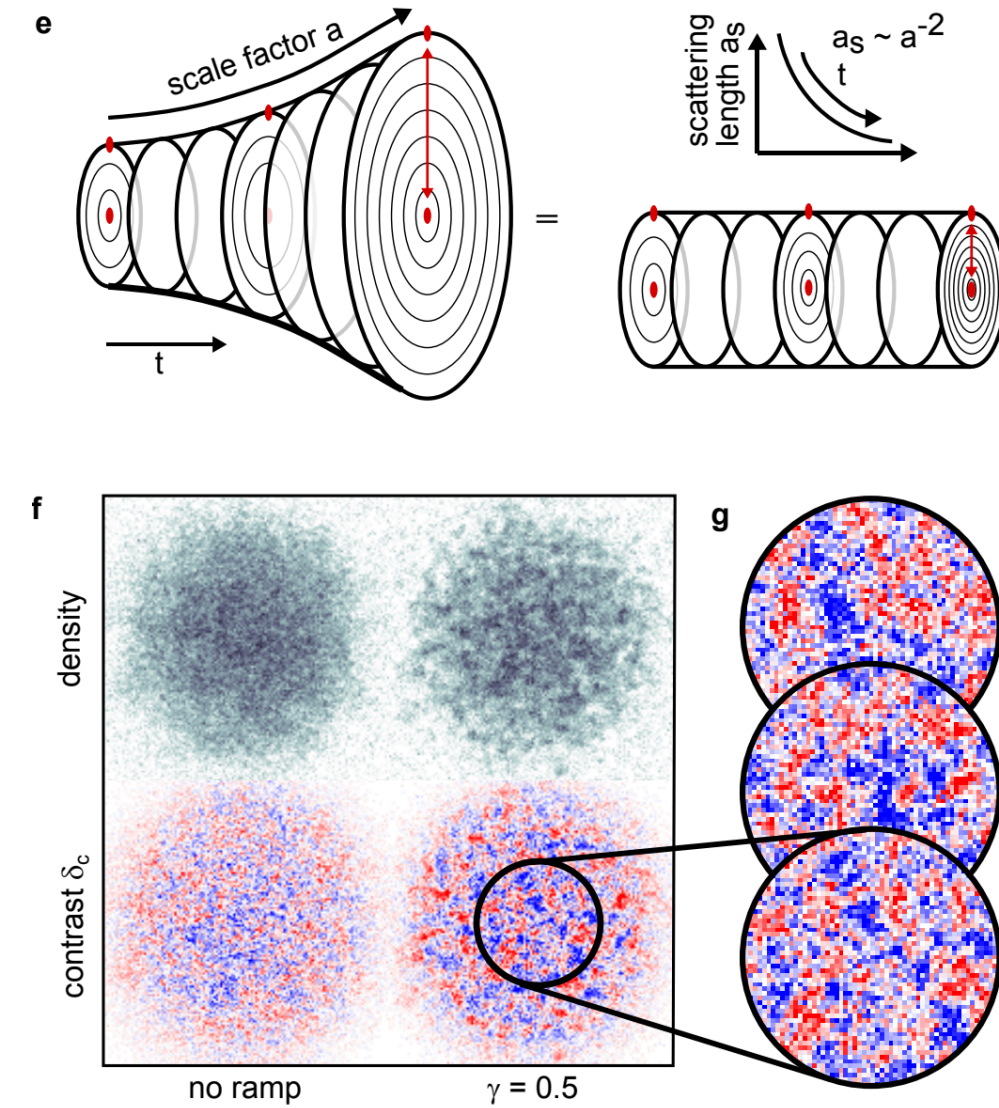


Analogue Black Hole and Hawking Radiation

Garay, L. J., Anglin, J. R., Cirac, J. I., & Zoller, P.
PRA, 63(2), 023611 (2001).

Steinhauer, J. *Nature Physics*, 12(10), 959-965 (2016).

⋮

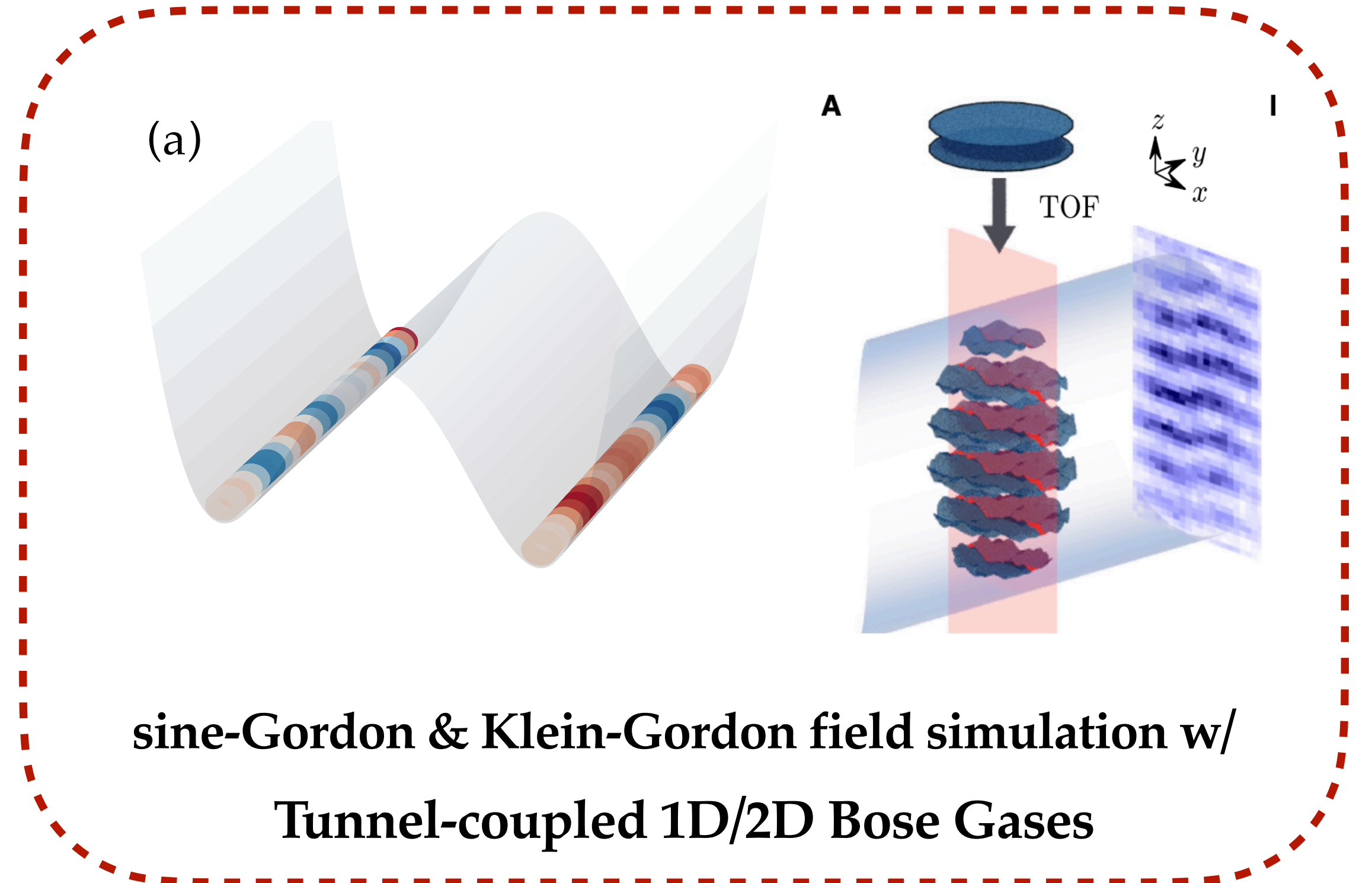


Quantum field in expanding spacetime

Viermann, C., et al. *Nature*, 611(7935), 260-264 (2022).

Agullo, I., Delhom, A., & Parra-López, Á. *PRD*, 110(12), 125023 (2024).

⋮



sine-Gordon & Klein-Gordon field simulation w/ Tunnel-coupled 1D/2D Bose Gases

Tajik, M., et al. *PNAS* 120(21), e2301287120 (2023).

Schweigler, T., et al. *Nature*, 545(7654), 323-326 (2017).

Sunami, S., et al. *PRL*, 128(25), 250402 (2022).

⋮

Outline

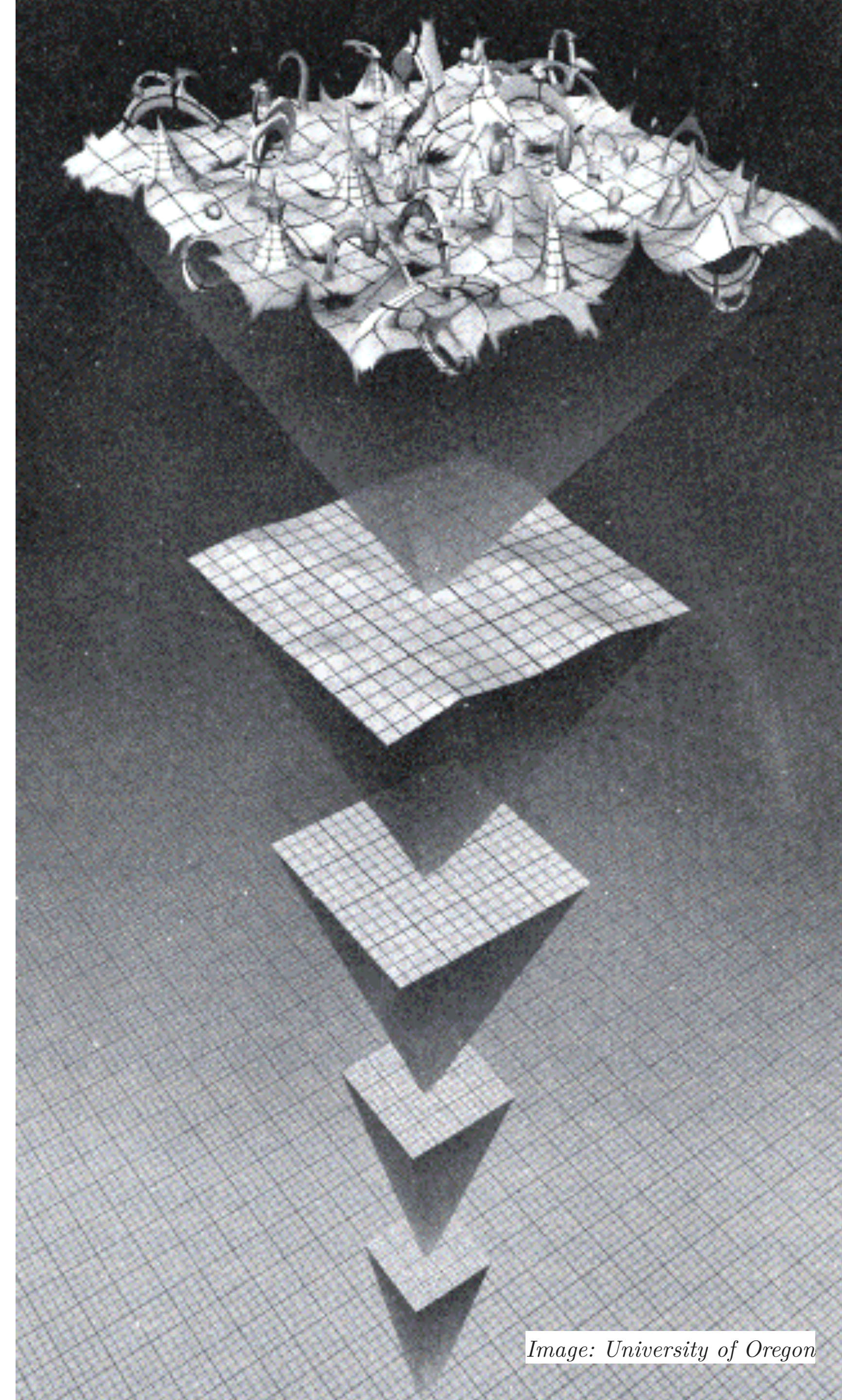
Motivation

Parallel 1D Bose Gases

Result: How to Measure Common Phase

Result: Field-space entanglement at finite temperatures

Open Questions and Outlook



Parallel 1D Bose Gases

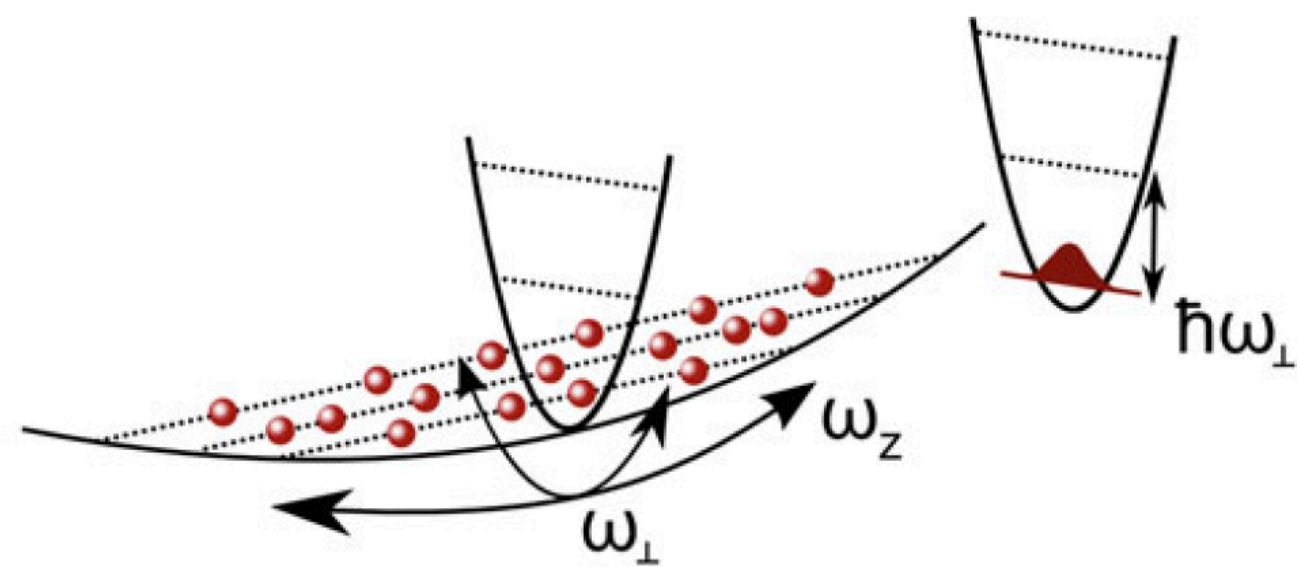
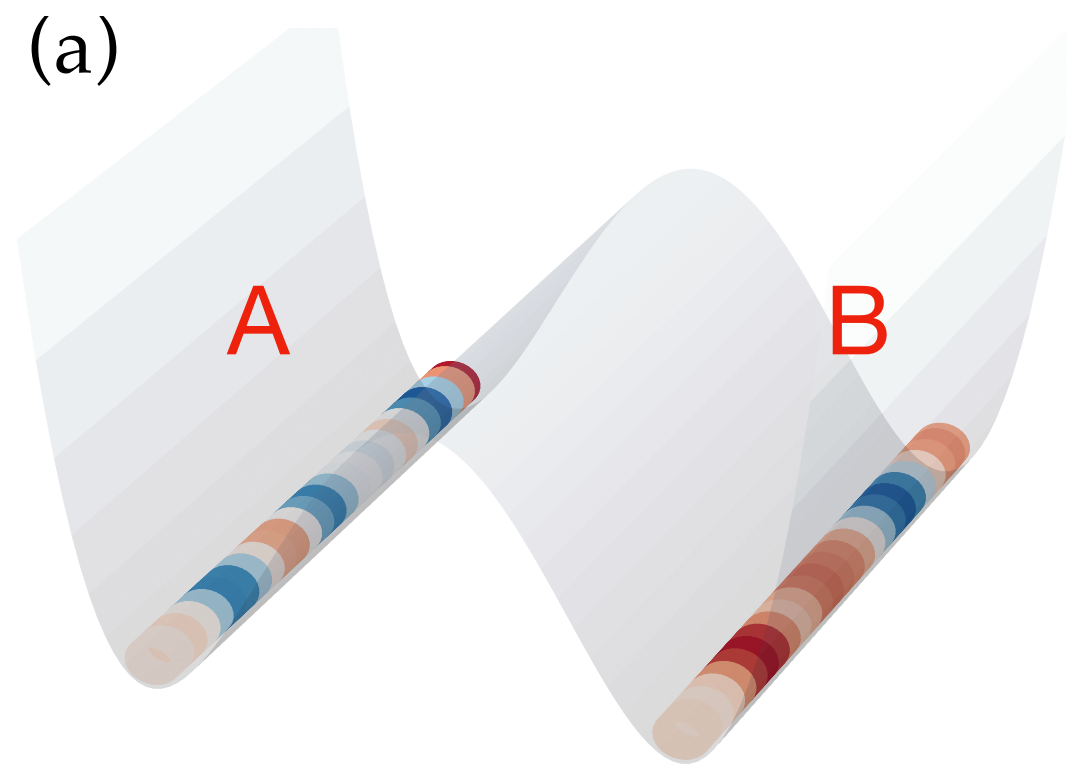


Image source: Tim Langen's PhD Thesis



Motional DoF of $\sim 10,000$ atoms
(collective), temperature: **20 - 100 nK**

Bosonic field operator $\hat{\Psi}_{A,B}(z)$ includes
both quantum and thermal fluctuations

$$[\hat{\Psi}_{\sigma}(z), \hat{\Psi}_{\kappa}(z')] = \delta(z - z')\delta_{\sigma\kappa}$$

Tunnel-coupled Lieb-Liniger model

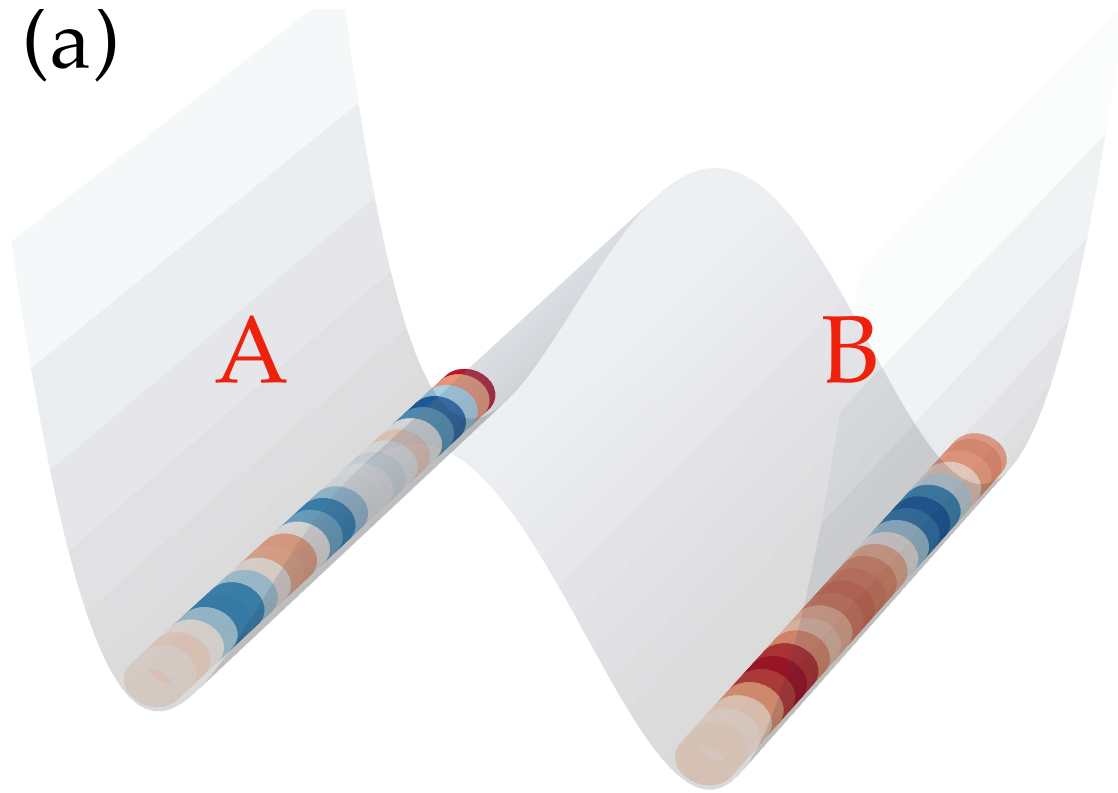
$$\hat{H} = \sum_{\sigma \in \{A,B\}} \left[\underbrace{-\frac{\hbar^2}{2m} \int \hat{\Psi}_{\sigma}^{\dagger} \partial_z^2 \hat{\Psi}_{\sigma}}_{\text{Kinetic + Potential}} + \underbrace{(V(z) - \mu) \int \hat{\Psi}_{\sigma}^{\dagger} \hat{\Psi}_{\sigma}}_{\text{Contact Interaction}} + \underbrace{\frac{g}{2} \int \hat{\Psi}_{\sigma}^{\dagger} \hat{\Psi}_{\sigma}^{\dagger} \hat{\Psi}_{\sigma} \hat{\Psi}_{\sigma}}_{\text{Contact Interaction}} \right] - \hbar J \int dz \left[\hat{\Psi}_A^{\dagger} \hat{\Psi}_B + \text{h.c.} \right]$$

Kinetic + Potential

Contact
Interaction

Tunnel-coupling

Effective Low-Energy Model



$$\hat{\Psi}_\sigma(z) = e^{i\hat{\phi}_\sigma(z)} \sqrt{n_0(z) + \delta\hat{n}_\sigma(z)}$$

$$[\delta\hat{n}_\sigma(z), \hat{\phi}_\kappa(z')] = i\delta(z - z')\delta_{\sigma\kappa}$$

Relative (-) and common (+) fields

$$\delta\hat{n}_\pm(z) = \delta\hat{n}_A(z) \pm \delta\hat{n}_B(z)$$

$$\hat{\phi}_\pm(z) = \hat{\phi}_A(z) \pm \hat{\phi}_B(z)$$

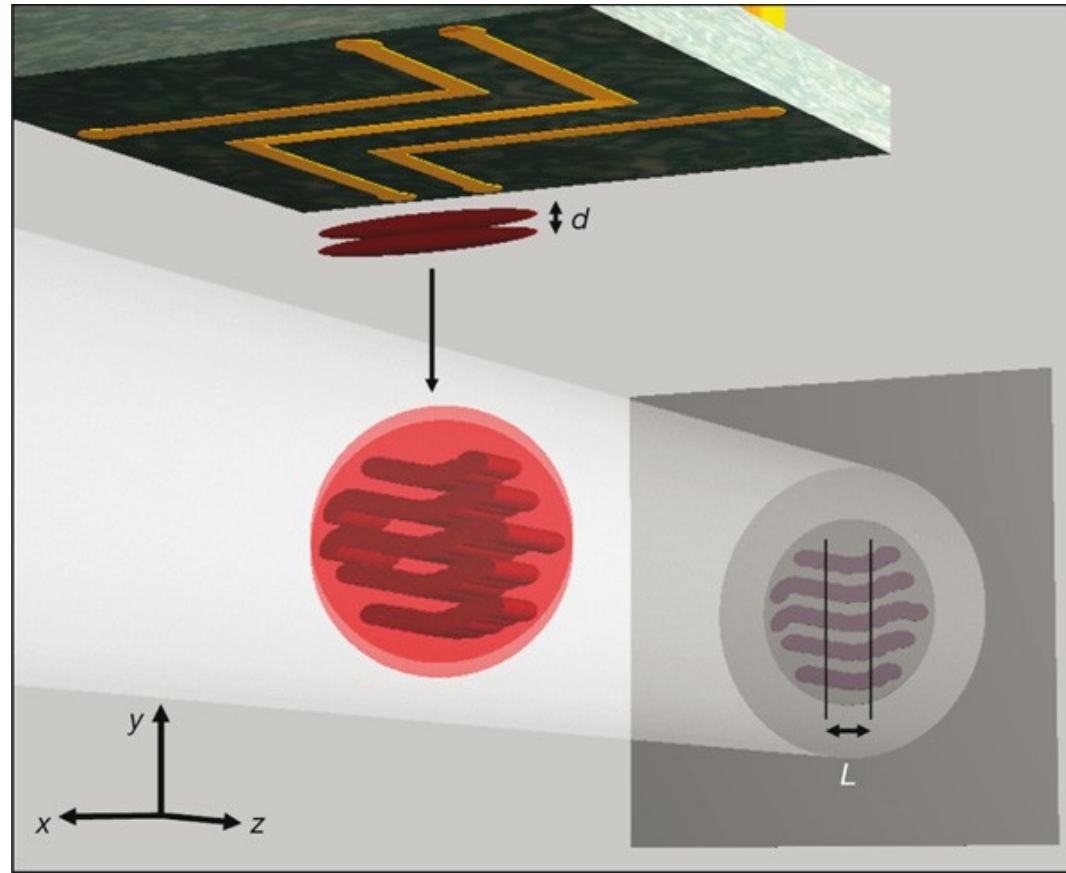
$$\hat{H} \approx \hat{H}_{SG} = \hat{H}_{TLL}[\delta\hat{n}_+, \hat{\phi}_+] + \hat{H}_{TLL}[\delta\hat{n}_-, \hat{\phi}_-] - 2\hbar J \int n_0(z) \cos \hat{\phi}_-(z) dz$$

$$\hat{H}_{TLL}[\delta\hat{n}, \hat{\phi}] = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}(z))^2 \right] dz$$

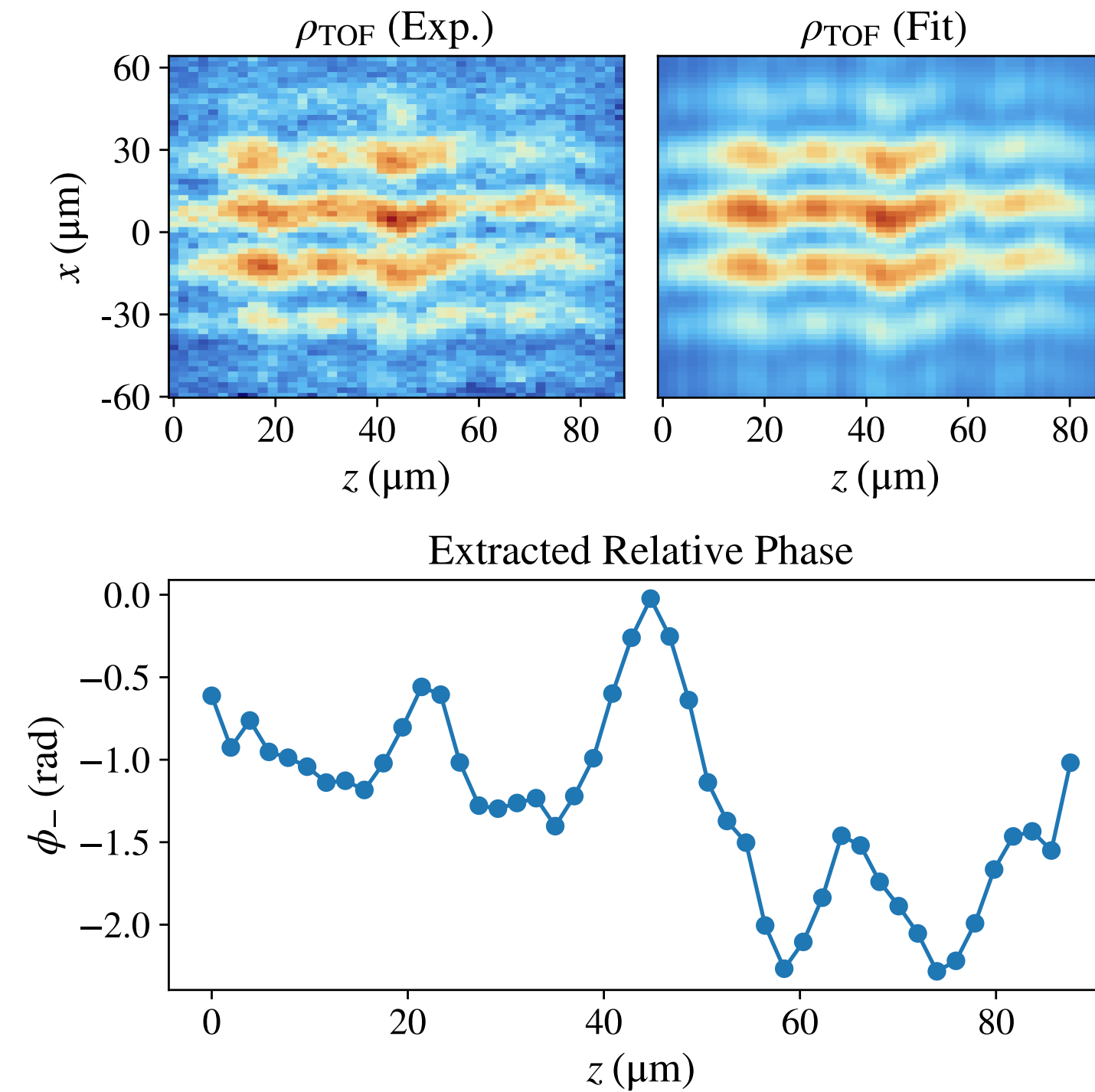
**sine-Gordon
interaction**

Tomonaga-Luttinger Liquid (TLL)

Matter-wave interference readout



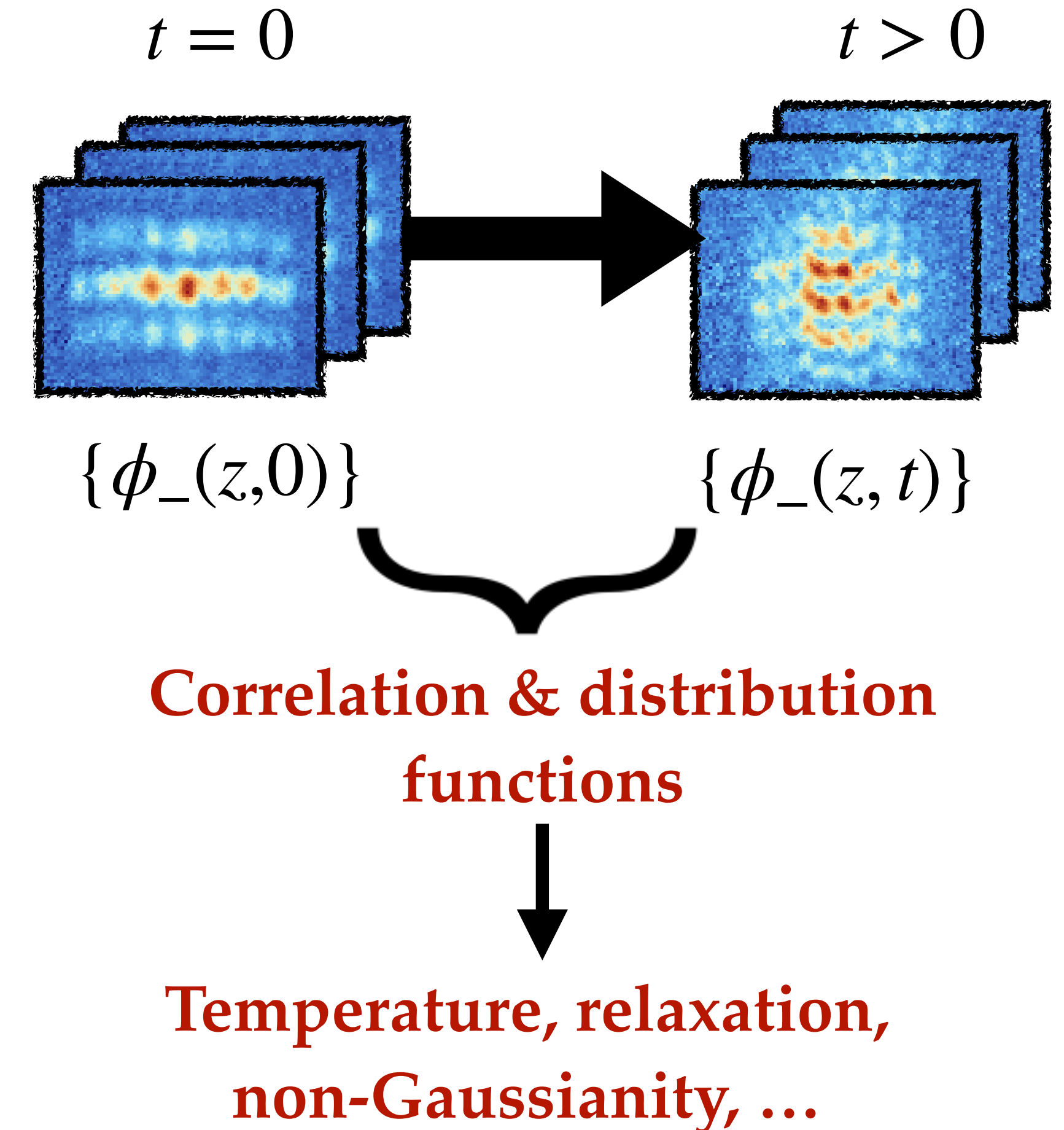
Hofferberth, S., Lesanovsky, I., Fischer, B., Schumm, T., & Schmiedmayer, J.
Nature, 449(7160), 324-327. (2007)



$$\rho_{\text{TOF}}(x, z, \tau) = A(z)e^{-x^2/\sigma_\tau^2} \left[1 + B(z) \cos(kx + \phi_-(z)) \right]$$

van Nieuwkerk, Y. D, Schmiedmayer, J. and Essler, F. *SciPost Physics* 5.5: 046 (2018).

Murtadho, T., et al. *SciPost Physics* 18.2: 065 (2025).

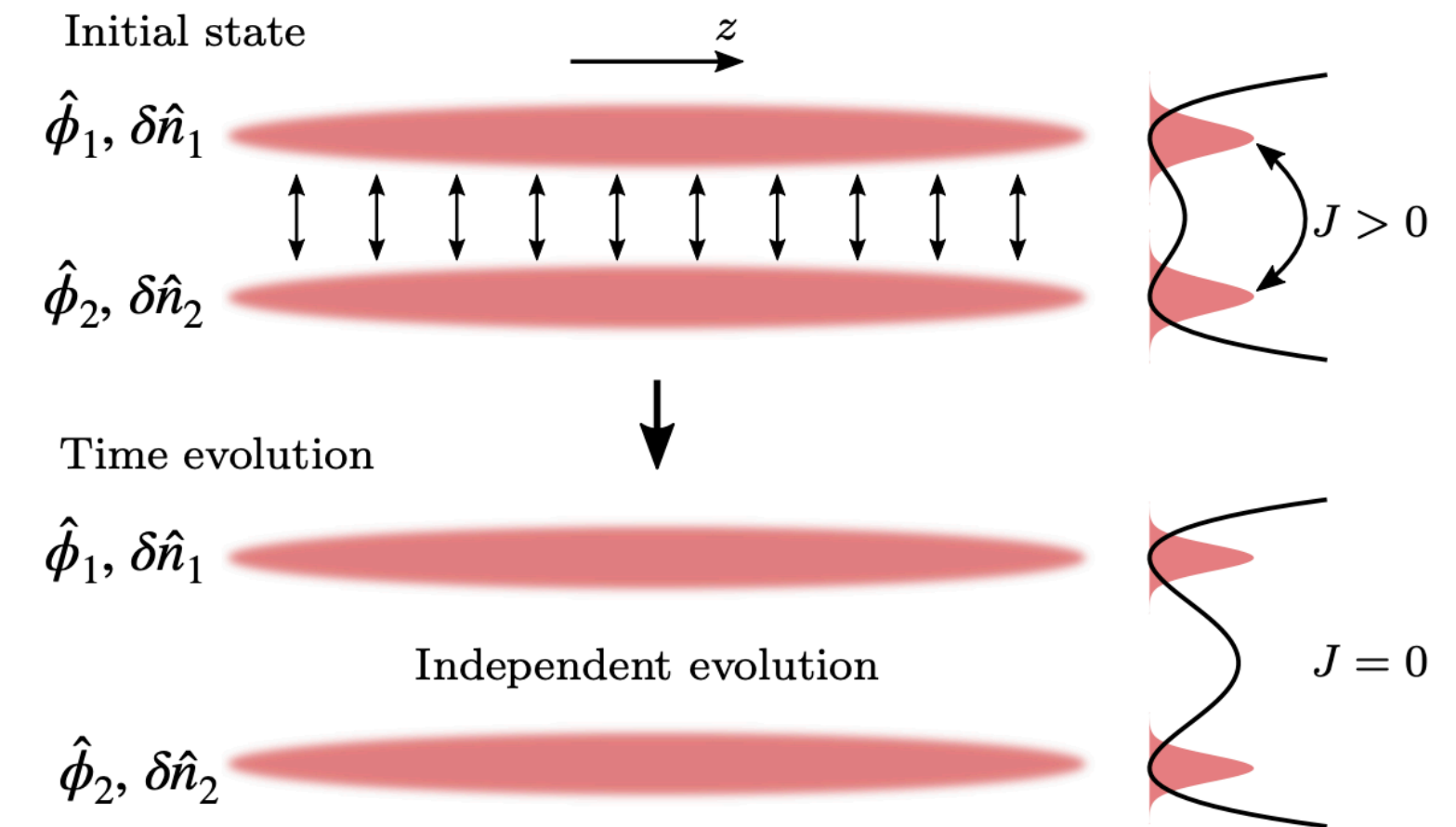


Gaussian Tomography

$$[\delta\hat{n}_-(z), \hat{\phi}_-(z')] = i\delta(z - z')$$

Gaussian state $\leftrightarrow$ **Covariance matrix** Γ

$$\delta\hat{n}_-(z) = \sum_{\mu} f_{\mu}(z) \delta\hat{n}_{-}^{\mu} \quad \hat{\phi}_-(z) = \sum_{\mu} f_{\mu}(z) \hat{\phi}_{-}^{\mu}$$



$$\Gamma^{-} = \begin{pmatrix} \Gamma_{\delta n \delta n}^{-} & \Gamma_{\delta n \phi}^{-} \\ \Gamma_{\phi \delta n}^{-} & \Gamma_{\phi \phi}^{-} \end{pmatrix}$$

$$\hat{H}_{\text{TLL}}^{-} = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}_-(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}_-(z))^2 \right] dz$$

TOF Measurement

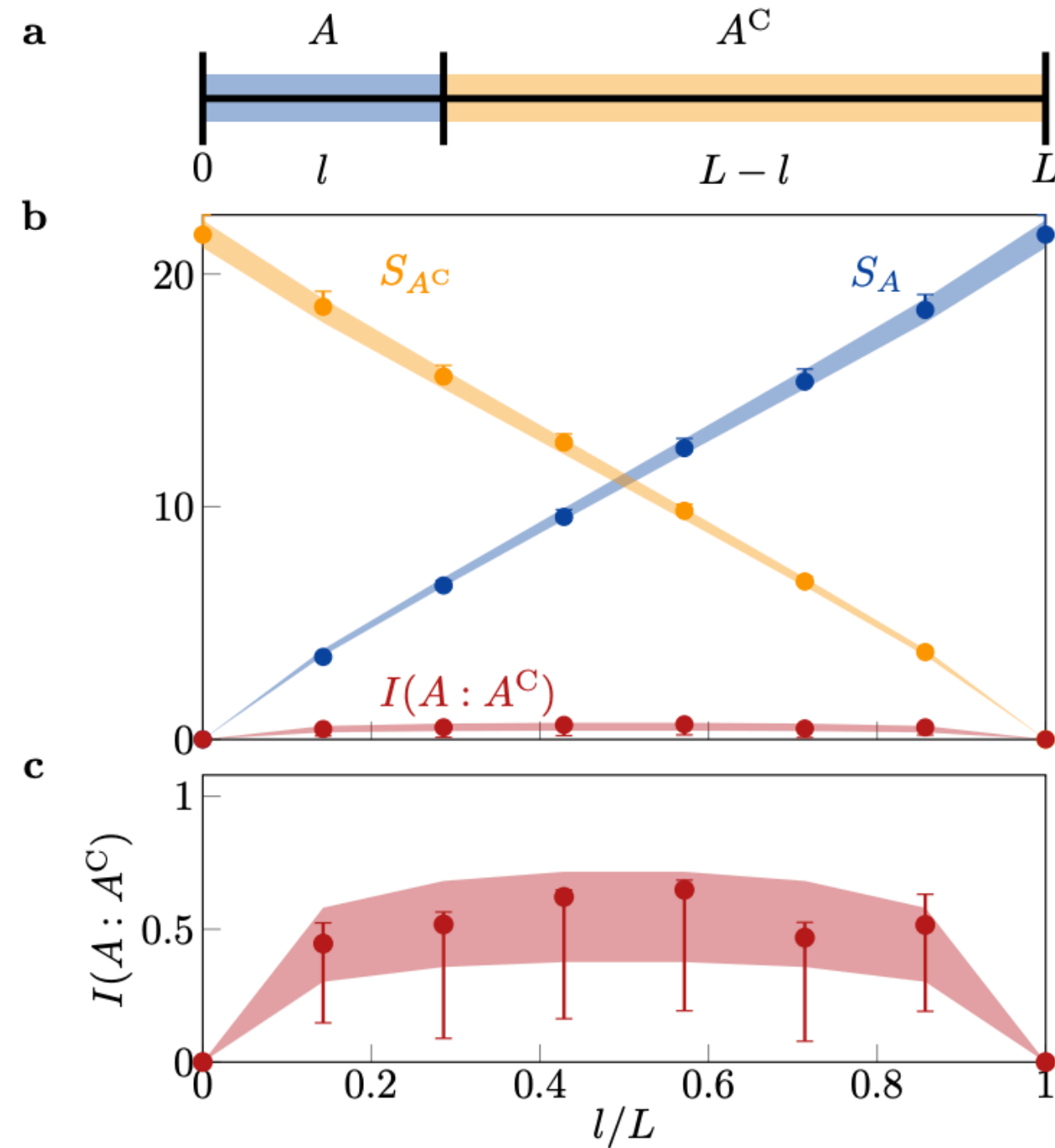
$$\{ \{ \phi_-(z, t_1) \}, \{ \phi_-(z, t_2) \}, \dots \}$$

Tomography

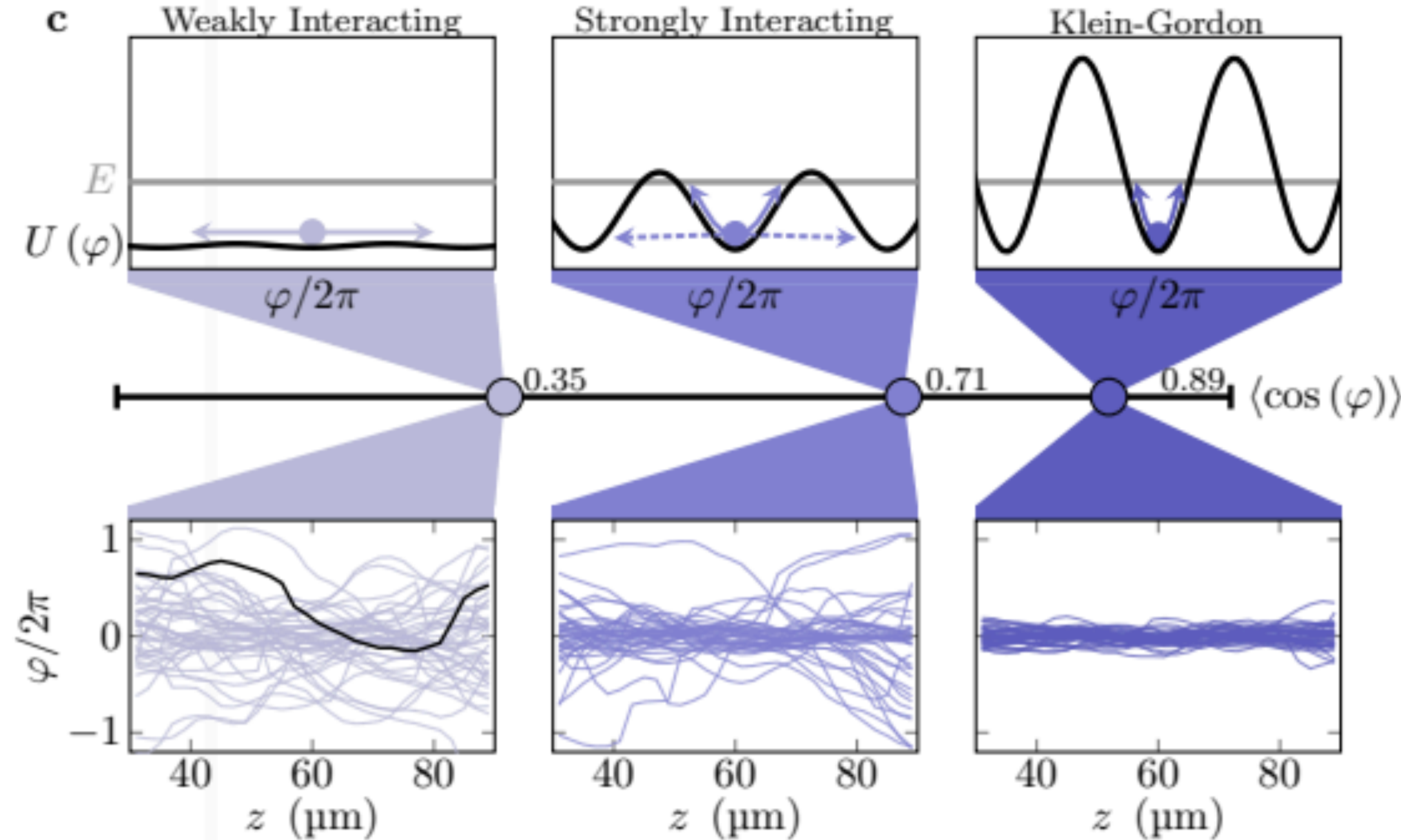
Gluza, M., et al. *Commun. Phys.* 3.1: 12 (2020).

Information-theoretic quantities: **entropies, mutual information,**

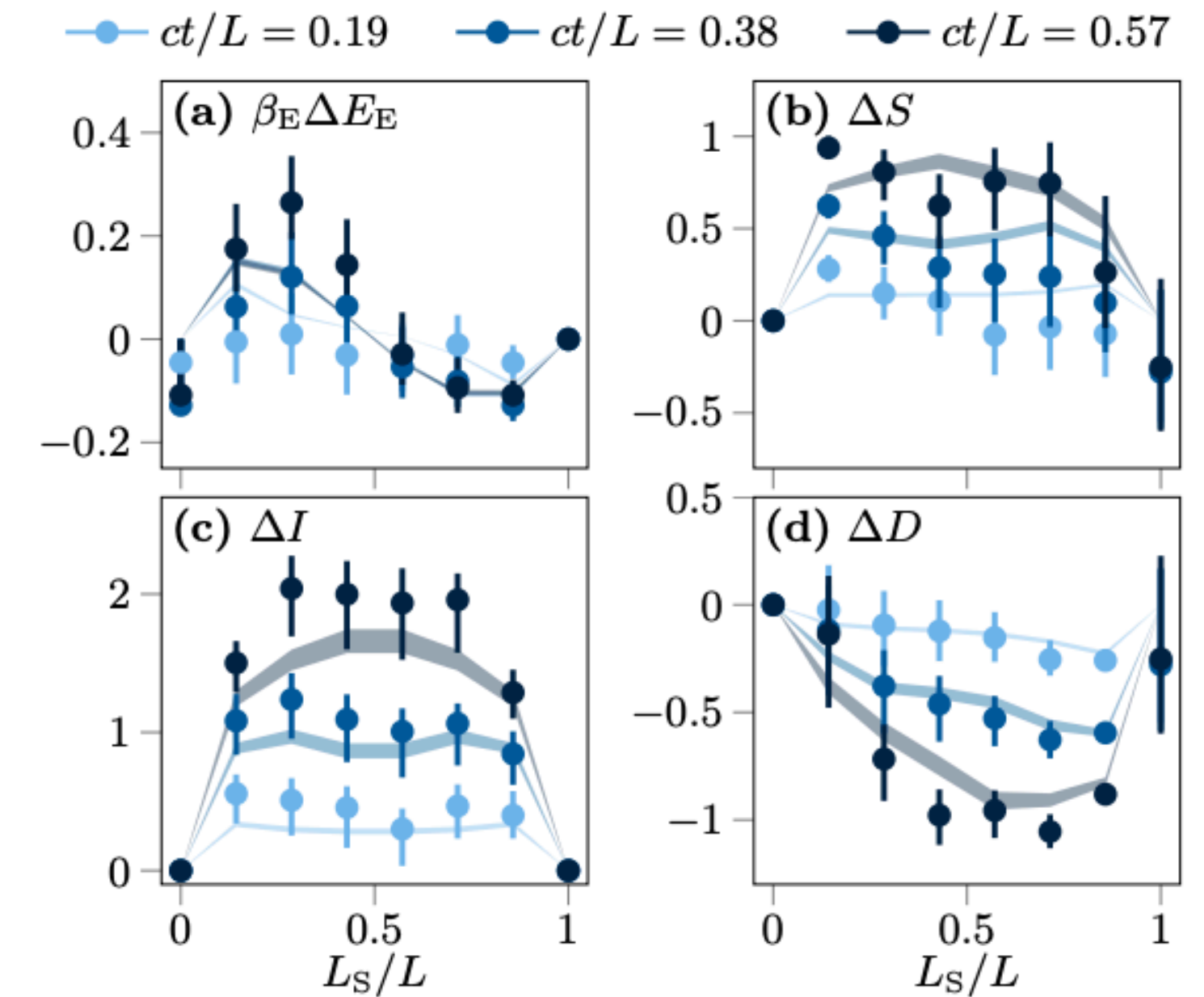
Field theory ft. information theory



Tajik, M., et al. *Nat. Phys.*, 19(7), 1022-1026 (2023).



Jarema, M. T., Tajik, M., Schmiedmayer, J., Weinfurtner, S., & Haas, T. *arXiv:2510.13783* (2025).



Aimet, S., et al. *Nat. Phys.*, 21(8), 1326-1331 (2025).

Entanglement??



Keep alive
quantum mechanics!

Measurement bottleneck

- * Limited available measurements create a bottleneck to the phenomena that can be observed
 - * Research aim: expand the measurement toolbox and use it to probe new physical phenomena (e.g. entanglement) in quantum field simulators
- 



Relative (-) and common (+) fields

$$\delta \hat{n}_{\pm}(z) = \delta \hat{n}_A(z) \pm \delta \hat{n}_B(z)$$

$$\hat{\phi}_{\pm}(z) = \hat{\phi}_A(z) \pm \hat{\phi}_B(z)$$

Outline

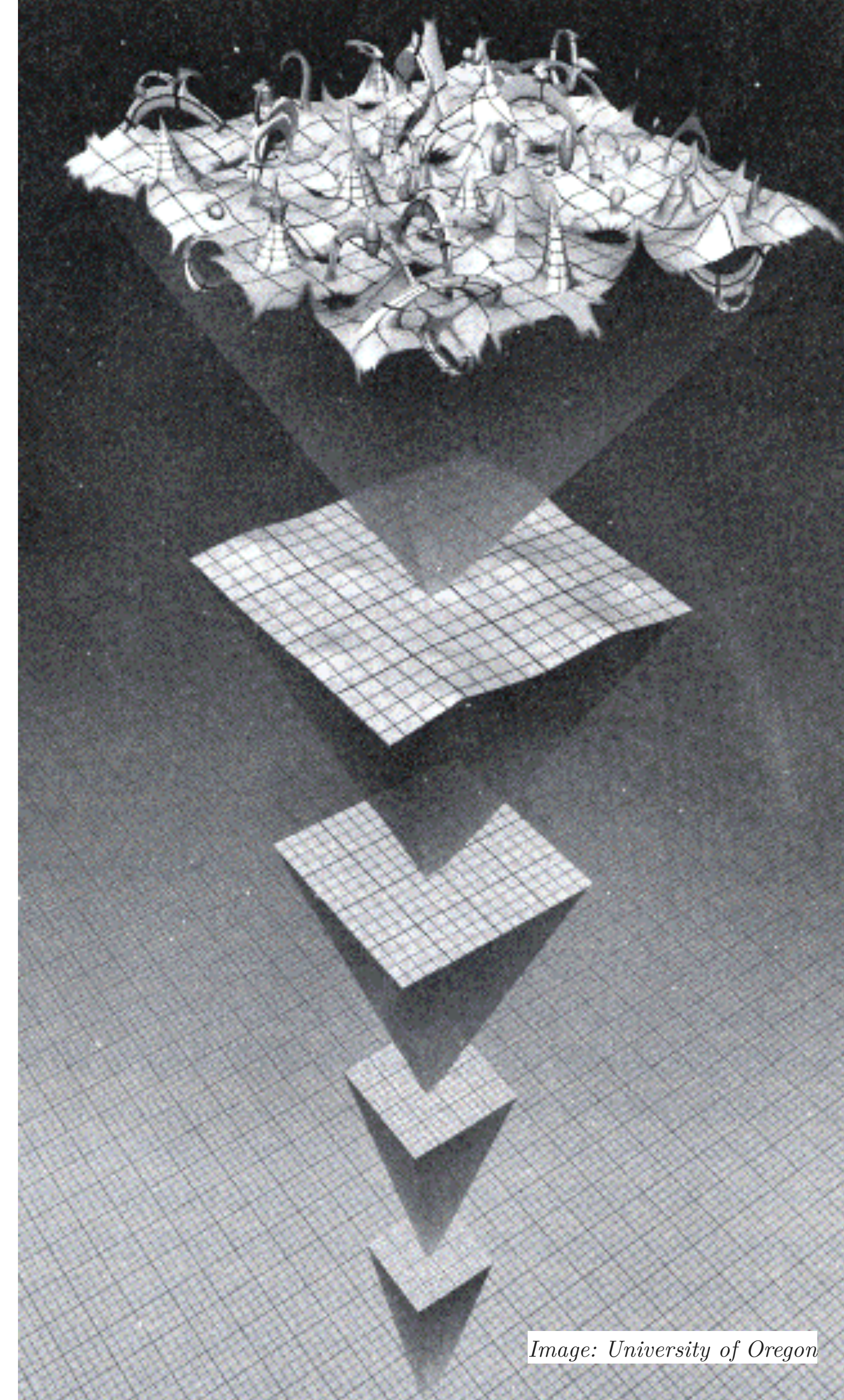
Motivation

Parallel 1D Bose Gases

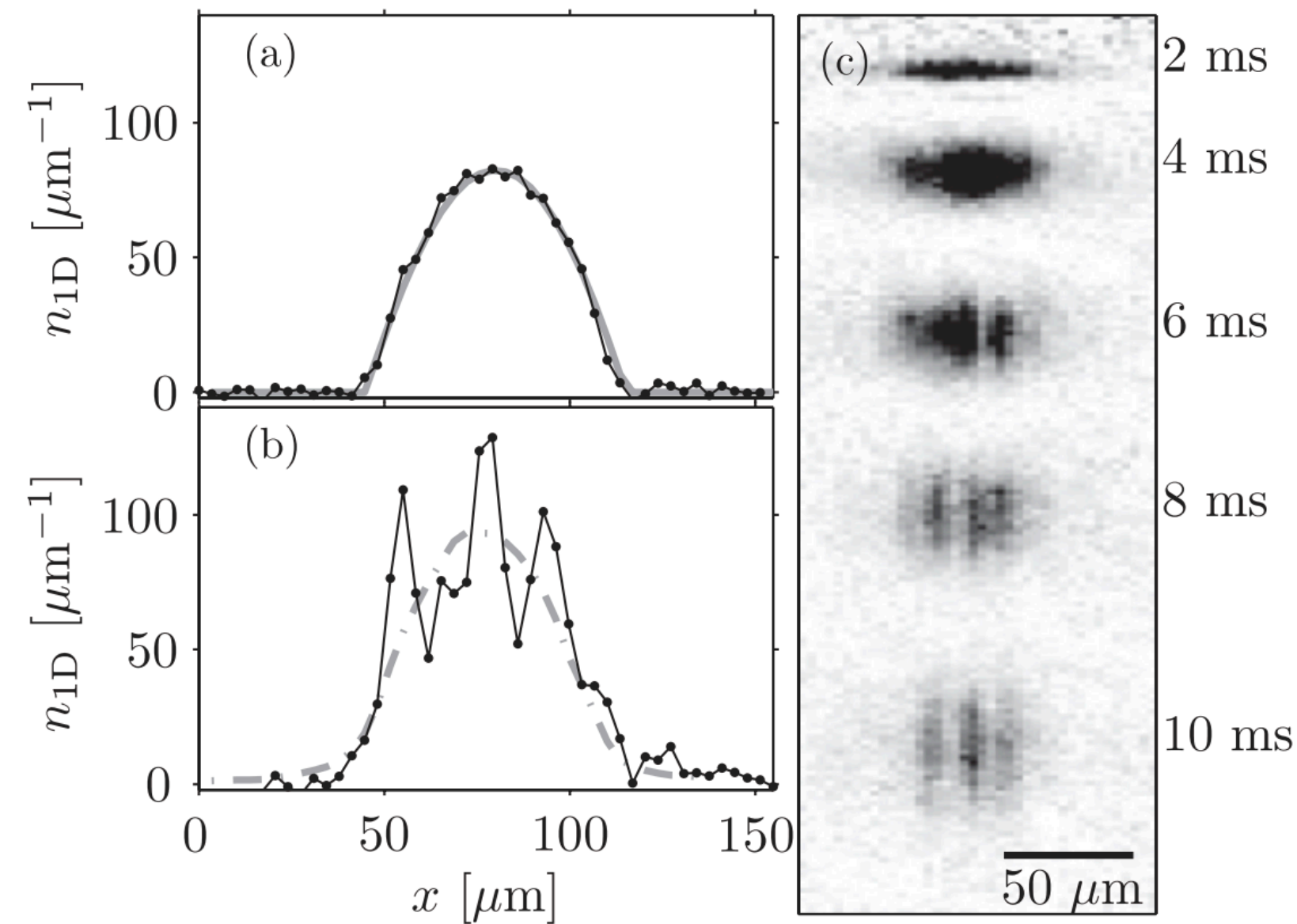
Result: How to Measure Common Phase

Result: Field-space entanglement at finite temperatures

Open Questions and Outlook



Extracting common phase

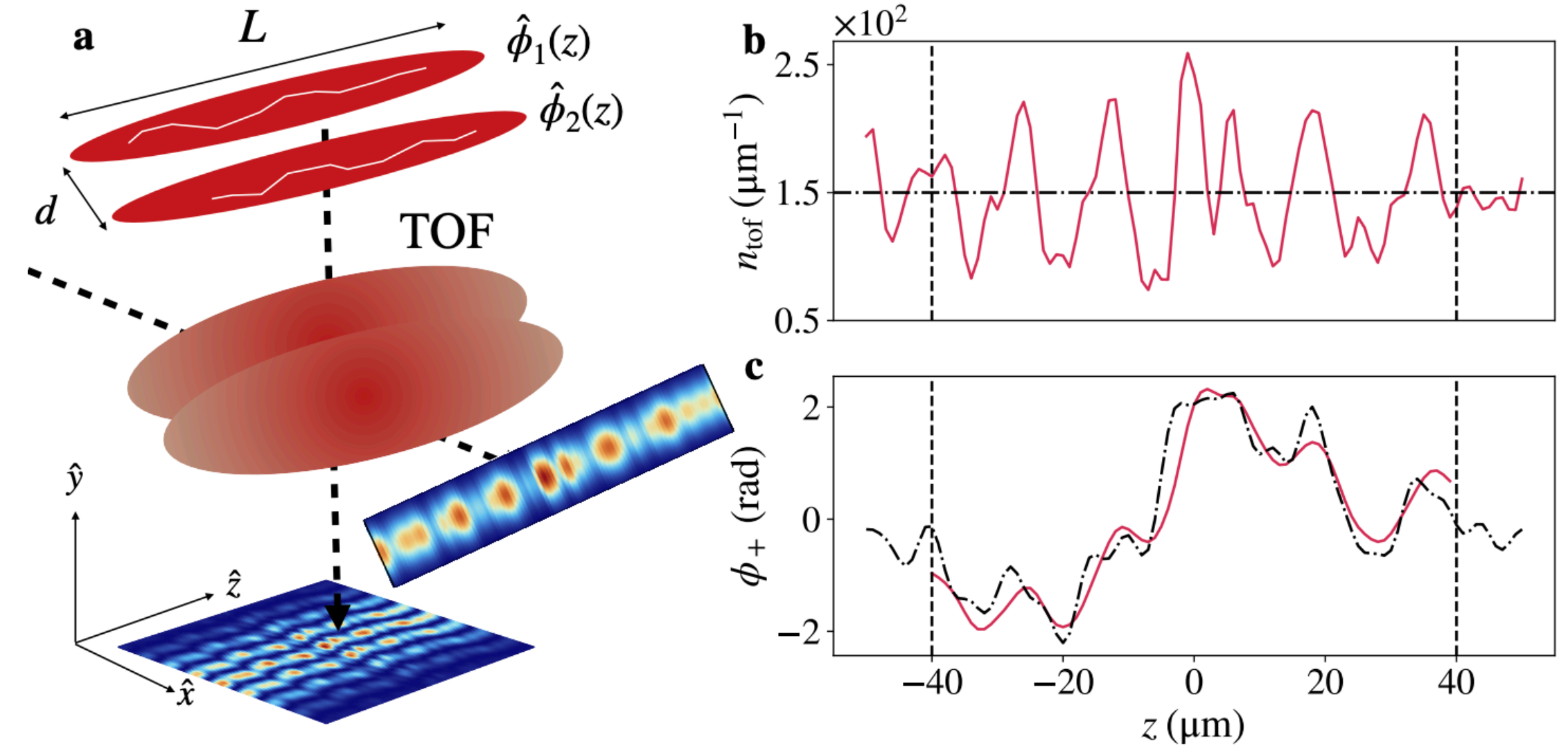


Manz, S., et al. *PRA*, 81(3), 031610 (2010).

Dettmer, S., et al. *PRL* 87(16), 160406 (2001).

$$\hat{\phi}(z) \rightarrow \delta\hat{n}(z, \tau)$$

$$\partial_\tau \hat{n} + \frac{\hbar}{2m} \nabla \cdot (n_0 \nabla \hat{\phi}) = 0$$



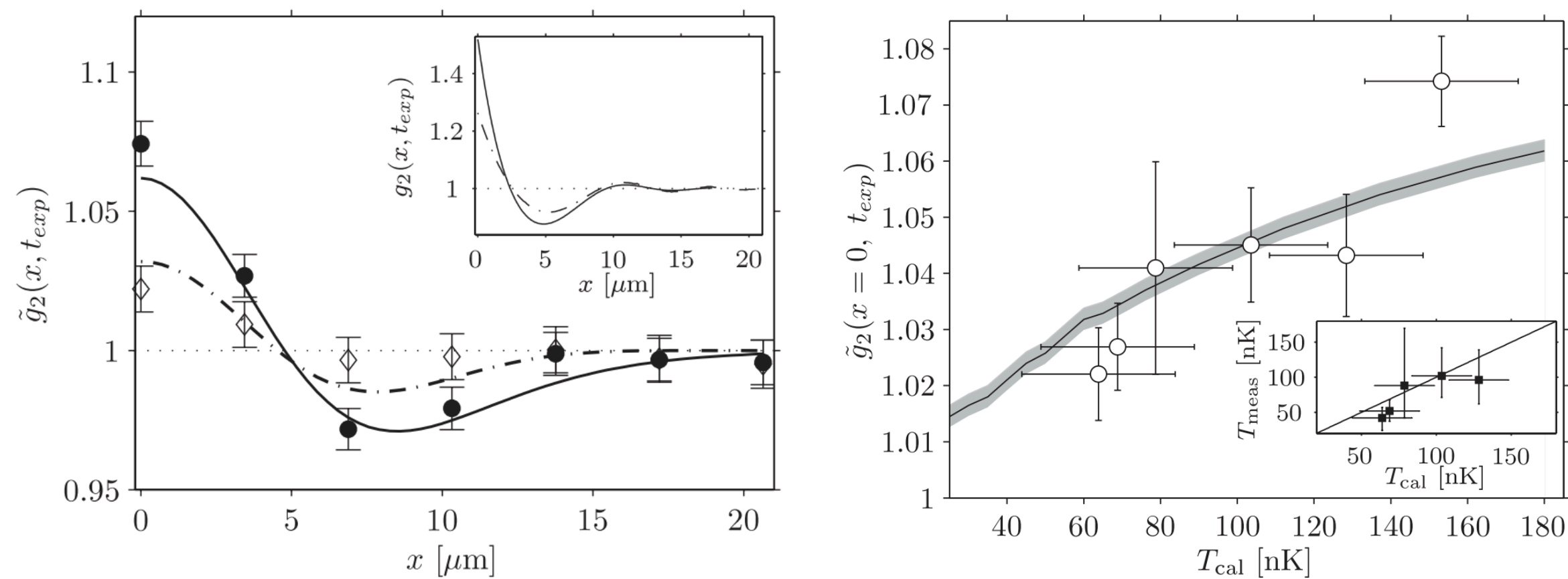
Murtadho, T., Cataldini, F., Erne, S., Gluza, M., Tajik, M., Schmiedmayer, J., & Ng, N. *PRR*, 7(2), L022031 (2025).

$$\hat{\phi}_+(z) = \hat{\phi}_1(z) + \hat{\phi}_2(z)$$

$$\partial_z^2 \hat{\phi}_+ \approx \frac{1}{\ell_\tau^2} \left(1 - \frac{n_{\text{tof}}(z, \tau)}{n_0(z)} \right)$$

Numerical Benchmarking (1)

Density ripple thermometry

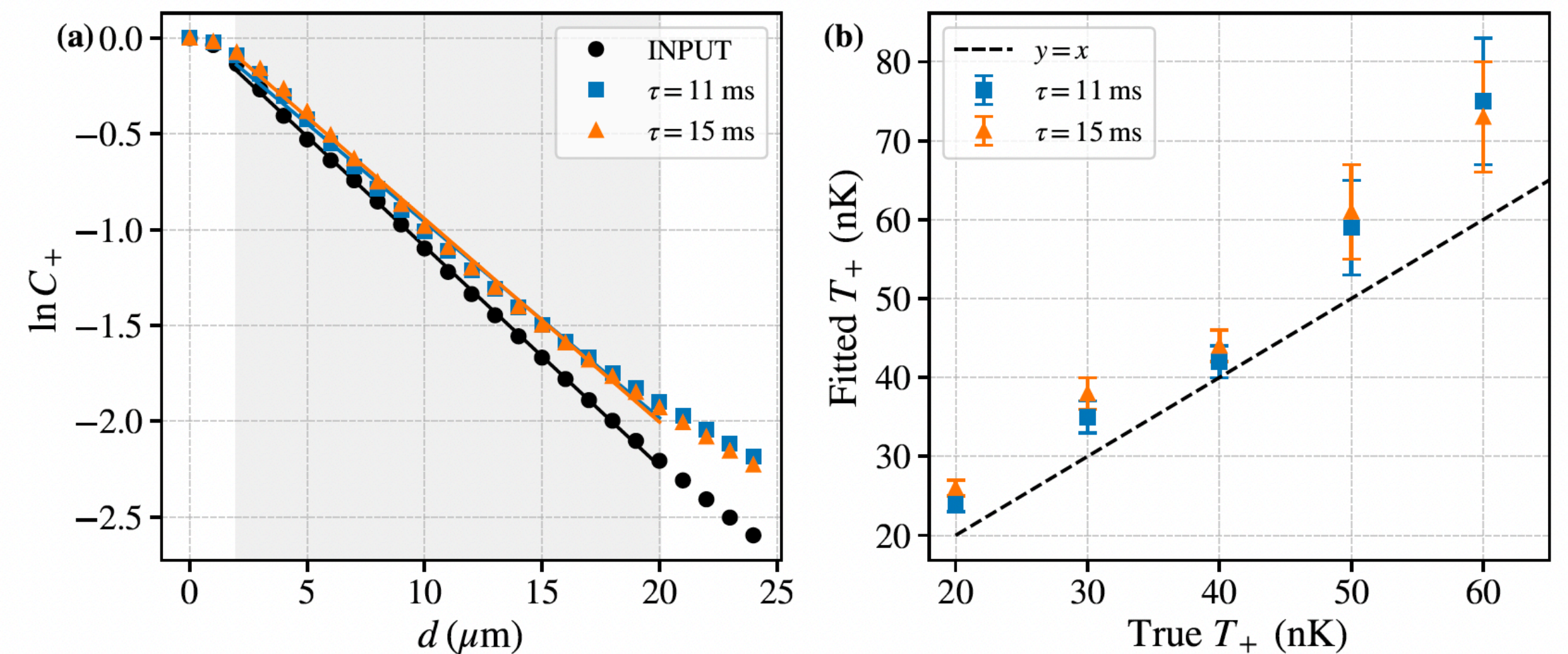


Imambekov, A., et al. *PRA*, 80(3), 033604 (2009).

Manz, S., et al. *PRA*, 81(3), 031610 (2010).

$$g_2(x, \tau) = \frac{\langle n(0, \tau) n(x, \tau) \rangle}{\langle n(0, \tau) \rangle \langle n(x, \tau) \rangle}$$

Vertex correlation thermometry

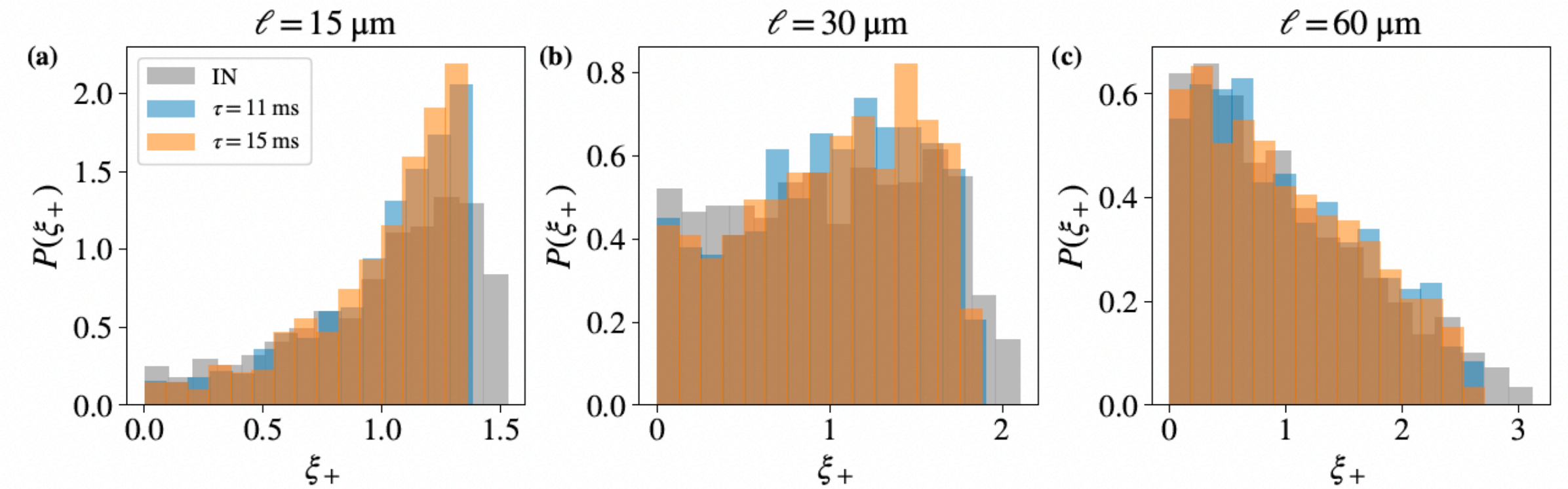
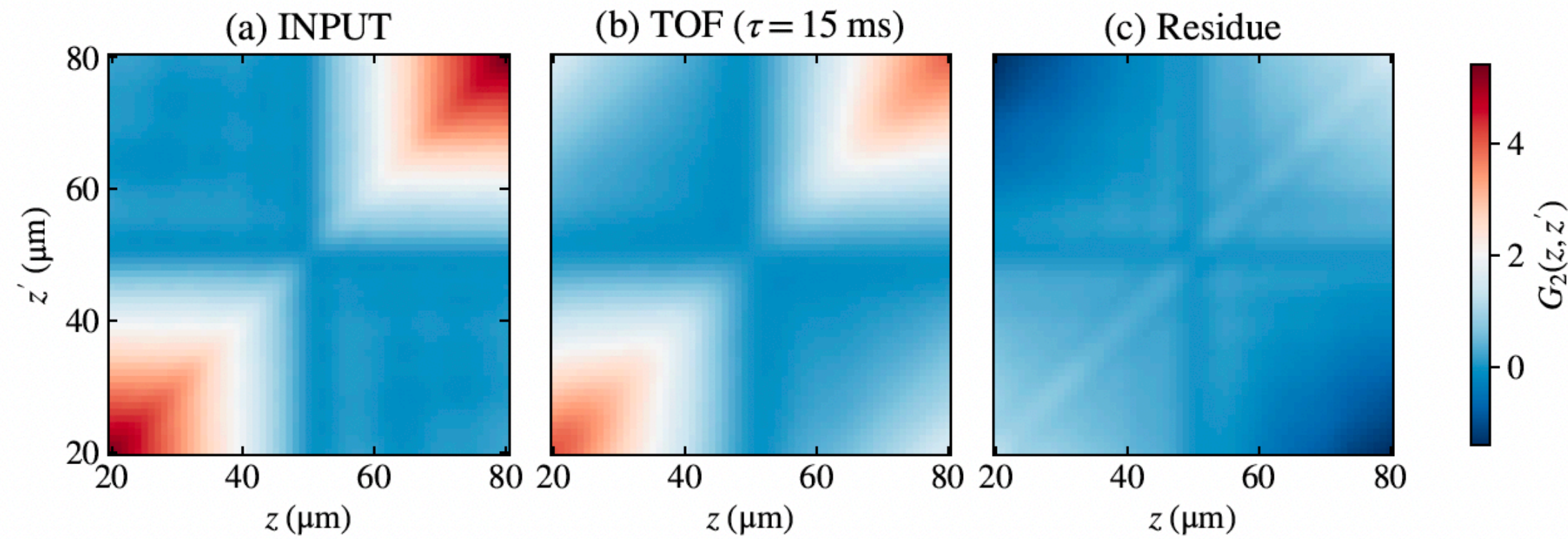


$$C_+(z, z') = \left\langle \exp [i(\phi_+(z) - \phi_+(z'))] \right\rangle$$

$$\sim \exp(-|z - z'|/\lambda_{T_+})$$

$$\lambda_{T_+} = \frac{\hbar^2 n_0}{mk_B T_+}$$

Numerical Benchmarking (2)



Contrast Distribution Function

$$\xi_+(\ell) = \frac{\left| \int_{-\ell/2}^{\ell/2} \exp(i\phi_+(z)) dz \right|^2}{\left\langle \left| \int_{-\ell/2}^{\ell/2} \exp(i\phi_+(z)) dz \right|^2 \right\rangle}$$

Two-point function

$$G_+^{(2)}(z, z') = \langle [\phi_+(z) - \phi_+(0)][\phi_+(z') - \phi_+(0)] \rangle$$

Can be generalized to higher-order, i.e. $G_+^{(N)}(\mathbf{z})$

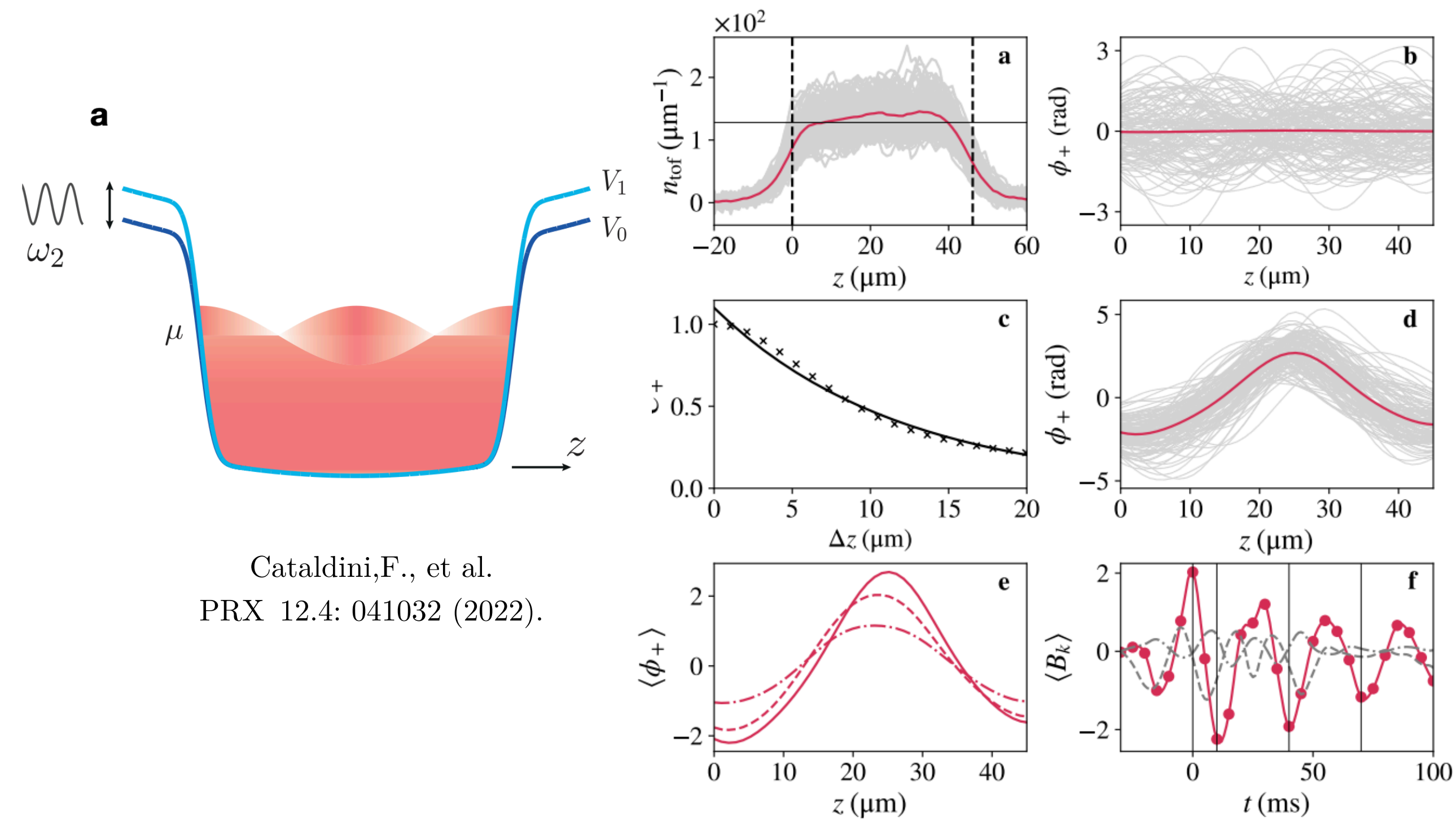
Gluza, M., et al. *Commun. Phys.* 3.1: 12 (2020).

Hofferberth, S., Lesanovsky, I., Schumm, T., Imambekov, A., Gritsev, V., Demler, E., & Schmiedmayer, J. *Nat. Phys.* , 4(6), 489-495 (2008).

All correlation functions probed for relative phase are now accessible for the common phase!

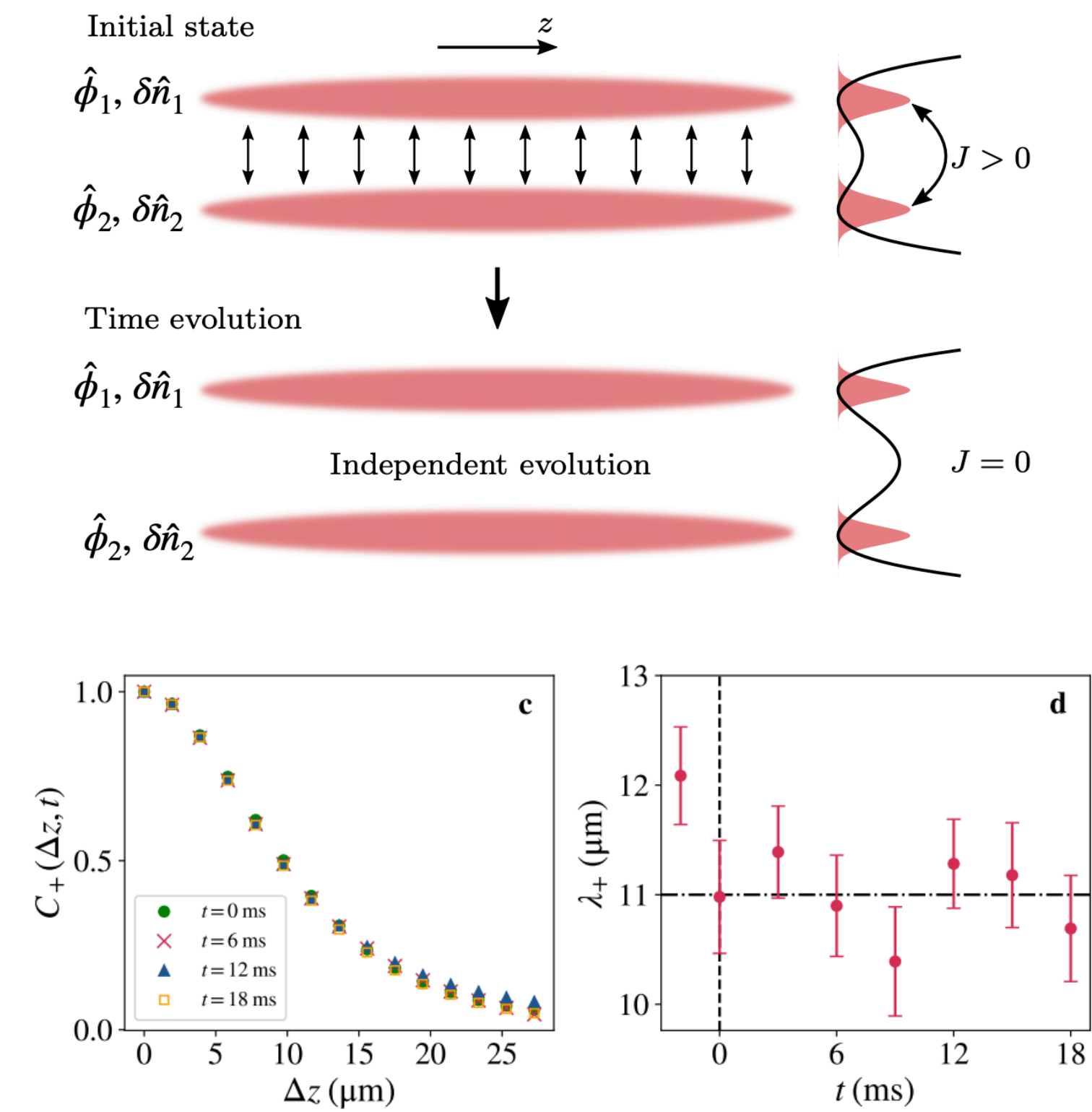
Experimental Benchmarking

Exp. 1: Relaxation of Driven Luttinger Liquids



Cataldini, F., et al.
PRX 12.4: 041032 (2022).

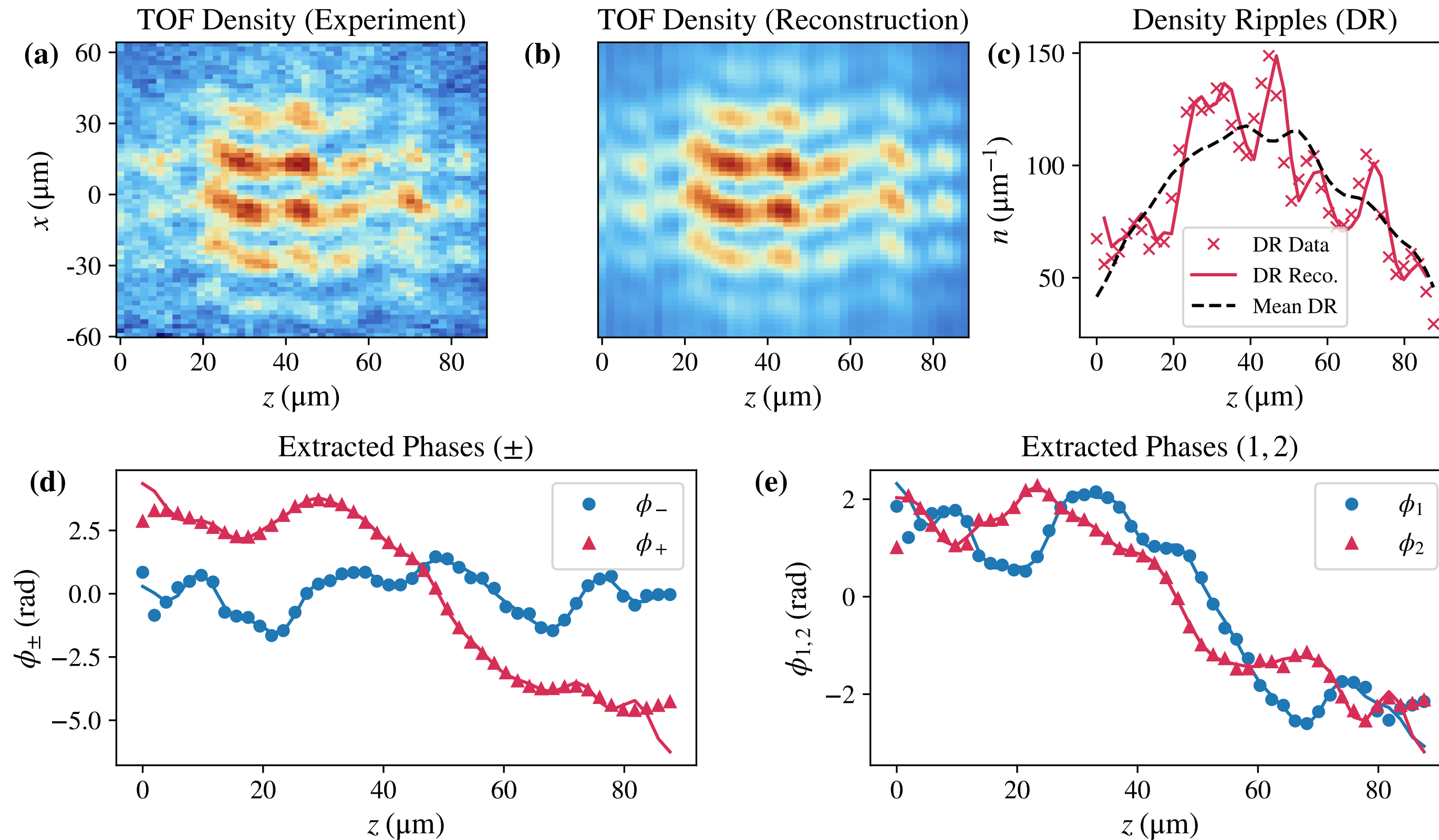
Exp. 2: Dynamics after tunneling quench



Schweigler, T., et al. *Nat. Phys.* 17.5: 559-563 (2021).

Probe $C_+(z, t)$
after quench
from $J > 0$ to
 $J = 0$

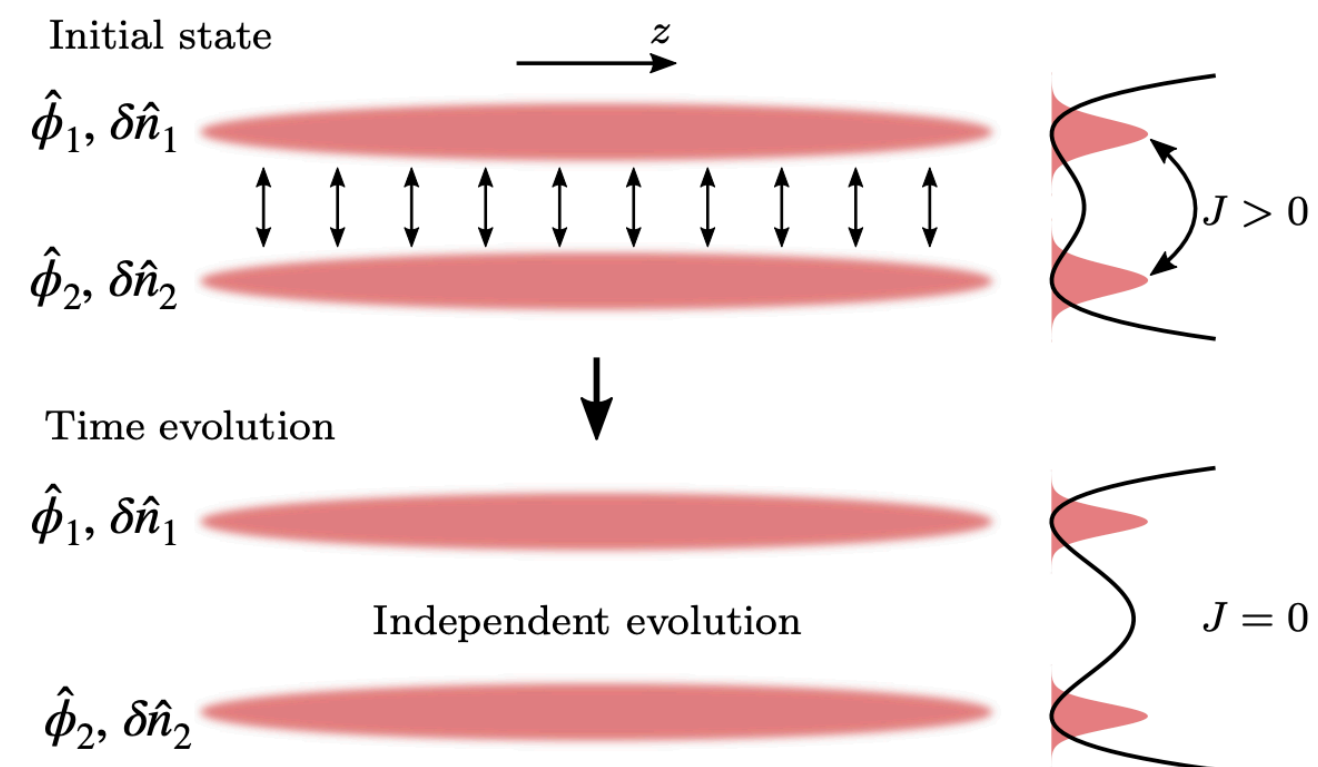
Expanding observable correlation functions



Direct access toward left-right correlation, e.g. $\langle \hat{\phi}_1(z) \hat{\phi}_2(z') \rangle$

$$\hat{\phi}_{1,2}(z) = \frac{1}{2}[\hat{\phi}_+(z) \pm \hat{\phi}_-(z)]$$

Full Gaussian Tomography



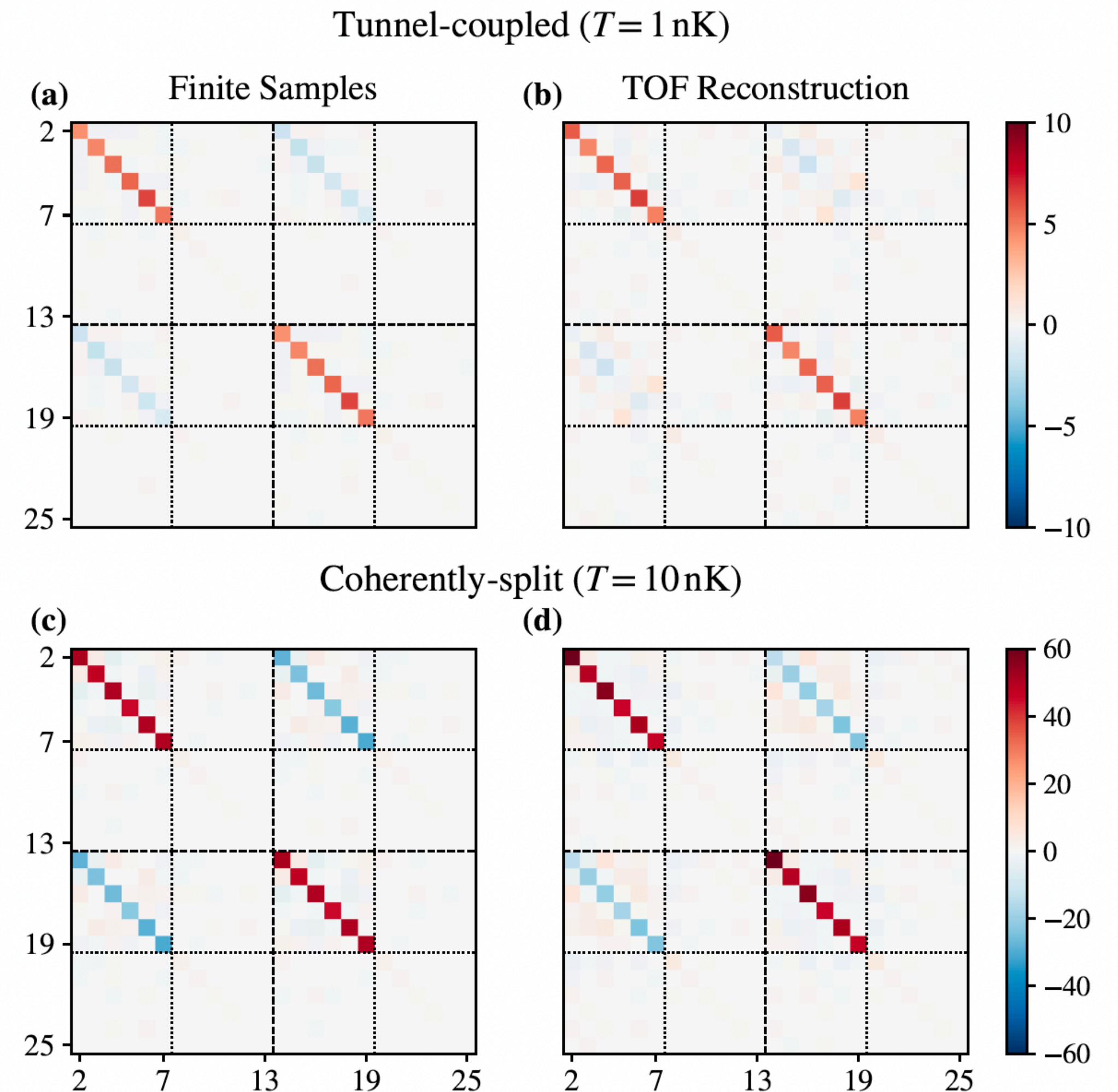
$$\hat{H}_{\text{TLL}}^{\pm} = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}_{\pm}(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}_{\pm}(z))^2 \right] dz$$

TOF Measurement

$$\{ \{ \phi_{\pm}(z, t_1) \}, \{ \phi_{\pm}(z, t_2), \dots \} \}$$

Full Tomography

$$\Gamma = \begin{pmatrix} \Gamma_{AA} & \Gamma_{AB} \\ \Gamma_{BA} & \Gamma_{BB} \end{pmatrix}$$



Outline

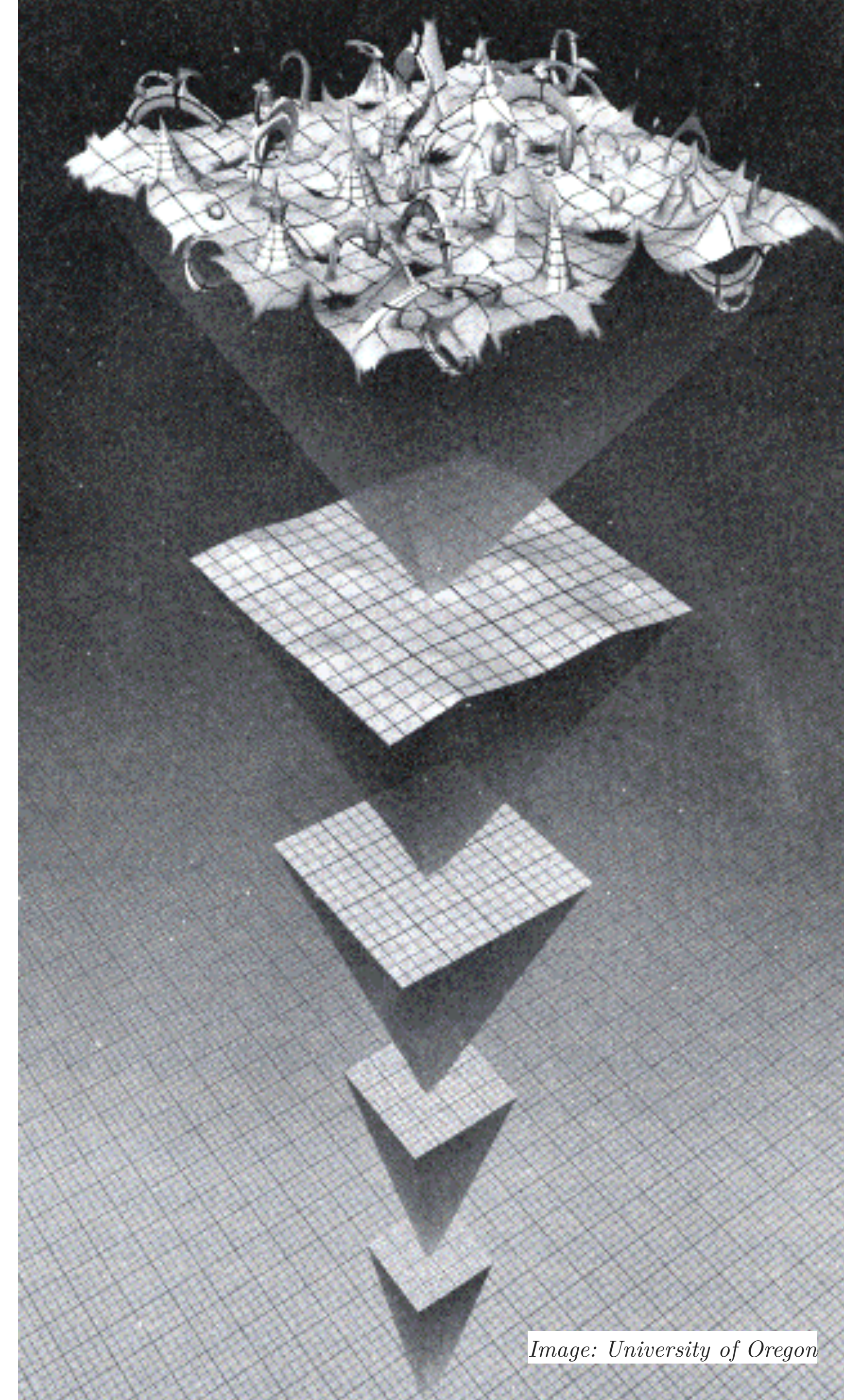
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Parallel 1D Bose Gases

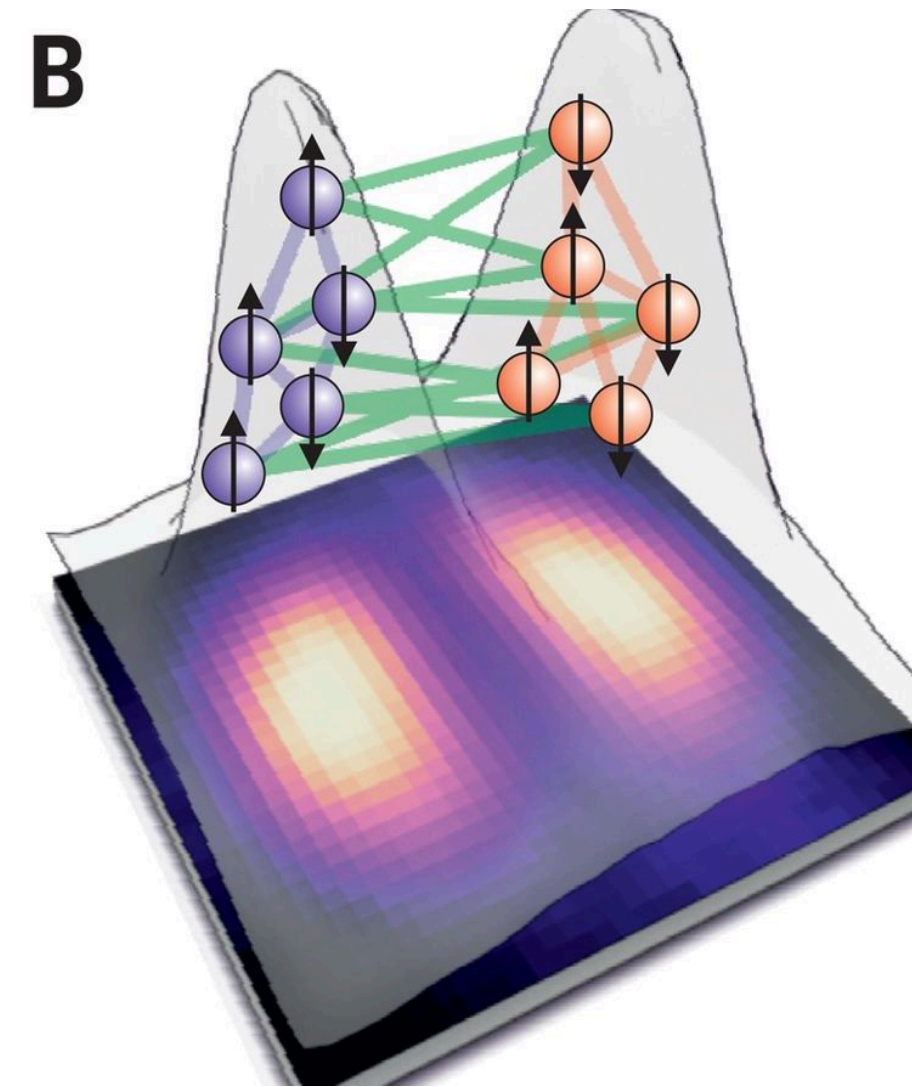
Result: How to Measure Common Phase

Result: Field-space entanglement at finite temperatures

Open Questions and Outlook

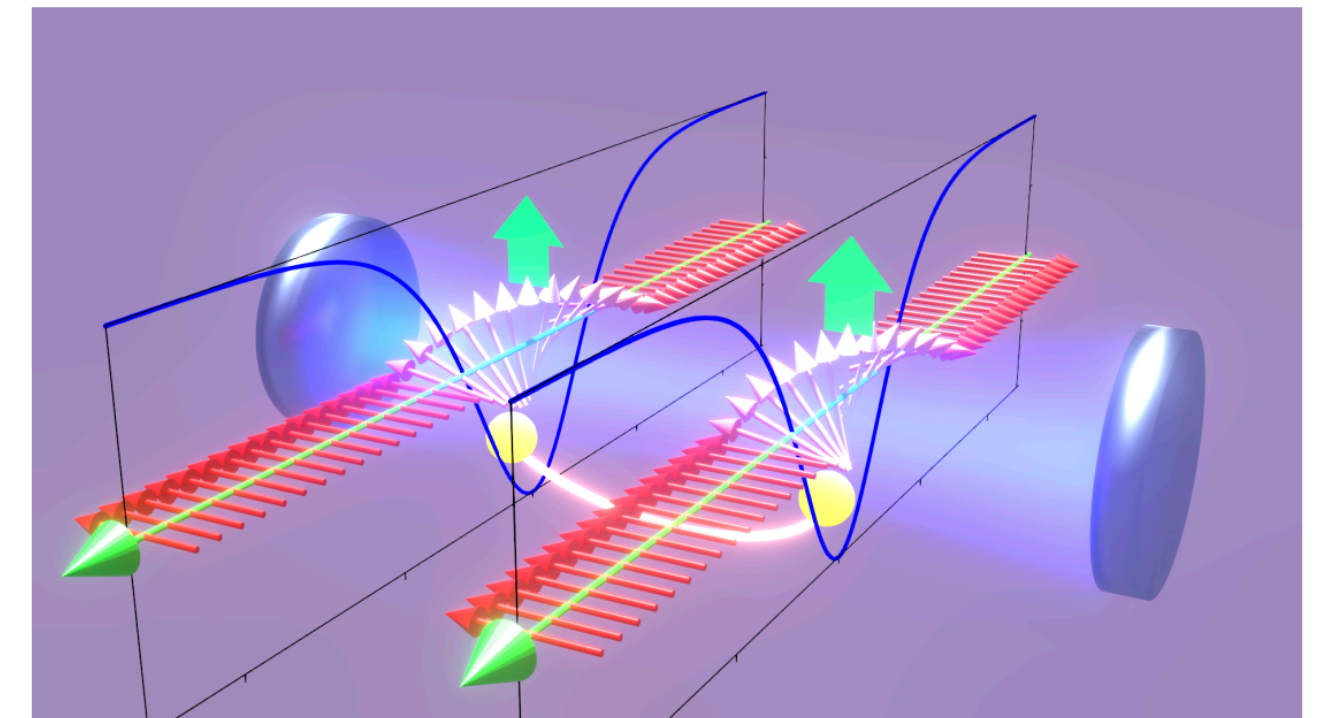


Entanglement: from micro to “macro” phenomenon



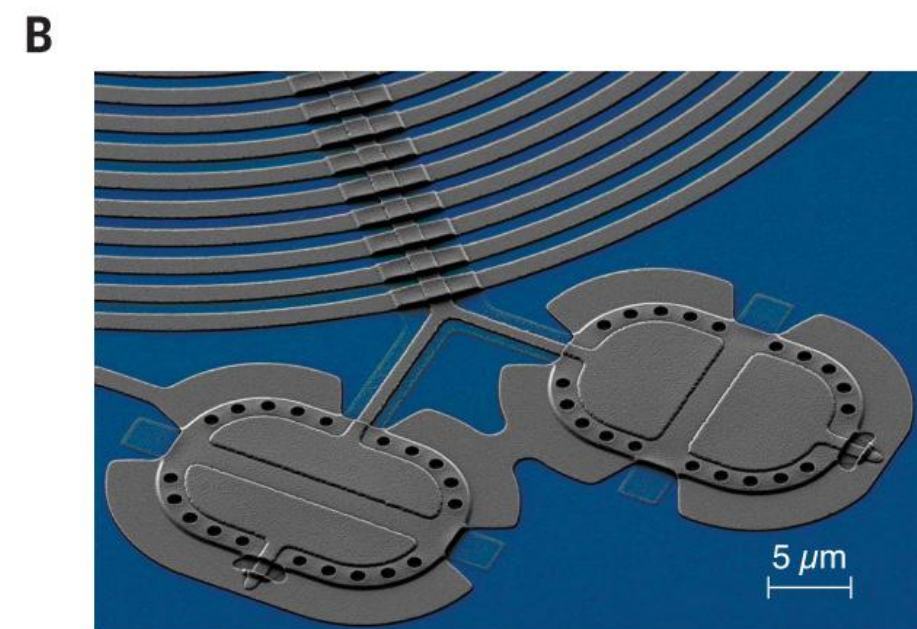
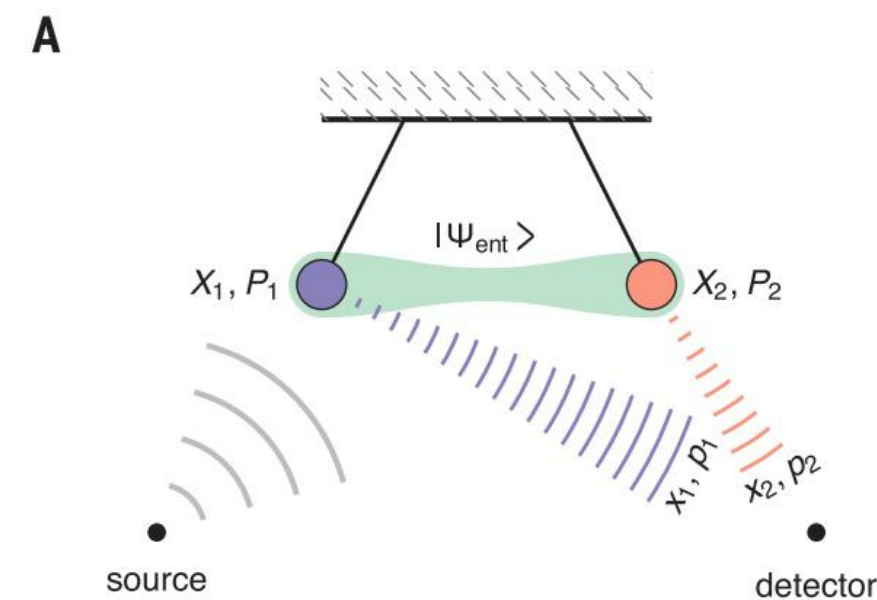
Macroscopic entanglement between spatially-separated spinor BECs

Lange, K., et al. *Science* 360.6387: 416-418 (2018).



Macroscopic entanglement between domain walls in optical cavity

Gupta, R., Yuan, H. Y., & Dhar, H. S.. *Phys. Rev. Appl.*, 24(6), 064067 (2025).

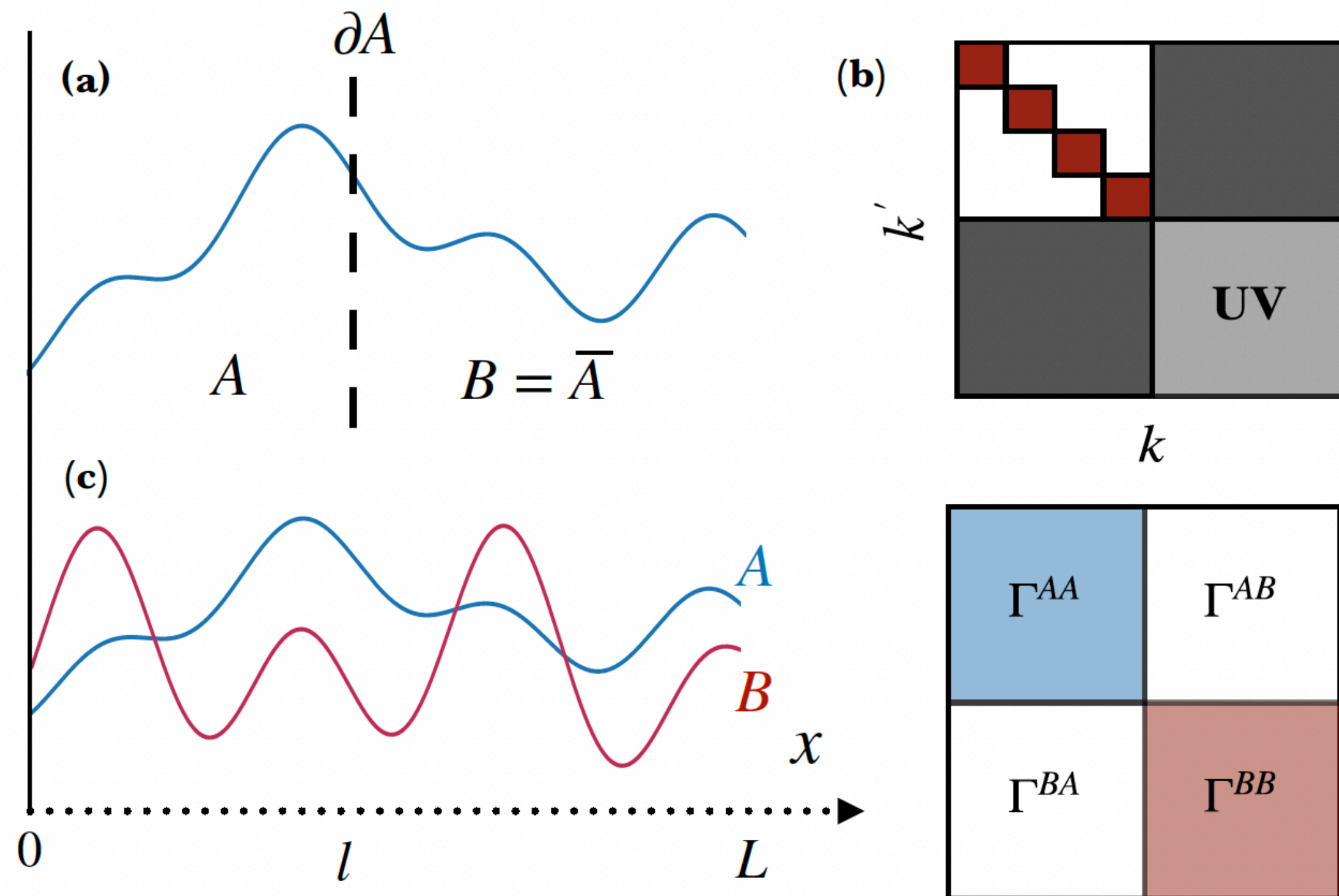


Macroscopic entanglement between mechanical oscillators

Kotler, S., et al. *Science* 372.6542: 622-625 (2021).

Alice & Bob of QFT

Entanglement depends on the **partition**



* Real-space entanglement

Area law of entanglement

Calabrese, P., and Cardy, J. JSTAT 2004.06: P06002 (2004).

.....

* Momentum-space entanglement

Balasubramanian, V., McDermott, M. B., & Van Raamsdonk, M.. PRD 86(4), 045014 (2012).

Kunkel, P. et al. PRL 128, 020402 (2022)

* Field-space entanglement

$$\hat{H} = \hat{H}_A[\hat{\phi}_A(x)] + \hat{H}_B[\hat{\phi}_B(x)] + \hat{H}_{\text{coupling}}[\hat{\phi}_A(x), \hat{\phi}_B(x)]$$

Mollabashi, A., Shiba, N., & Takayanagi, T. JHEP 2014(4), 1-36 (2014).

Review with Stefan, Eisert, et al. coming

Field's Covariance Matrix

Thermal density matrix

$$\hat{\rho}_{\text{th}} = \frac{1}{Z} \exp(-\beta \hat{H})$$

Collection of two-point correlation functions e.g. $\langle \hat{\phi}_k^+ \hat{\phi}_k^+ \rangle, \dots$



Covariance matrix

$$\Gamma = \begin{pmatrix} \Gamma^+ & 0 \\ 0 & \Gamma^- \end{pmatrix}$$

Symplectic transform.

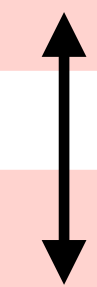
$$\tilde{\Gamma} = U \Gamma U^T$$

$$U(\Omega \oplus \Omega)U^T = \Omega \oplus \Omega$$

$$\tilde{\Gamma} = \begin{pmatrix} \Gamma^{AA} & \Gamma^{AB} \\ \Gamma^{BA} & \Gamma^{BB} \end{pmatrix}$$

Heisenberg uncertainty relation

$$\Gamma + i(\Omega \oplus \Omega) \geq 0 \quad \Omega = \begin{pmatrix} 0 & \mathbf{1}_\Lambda \\ -\mathbf{1}_\Lambda & 0 \end{pmatrix}$$

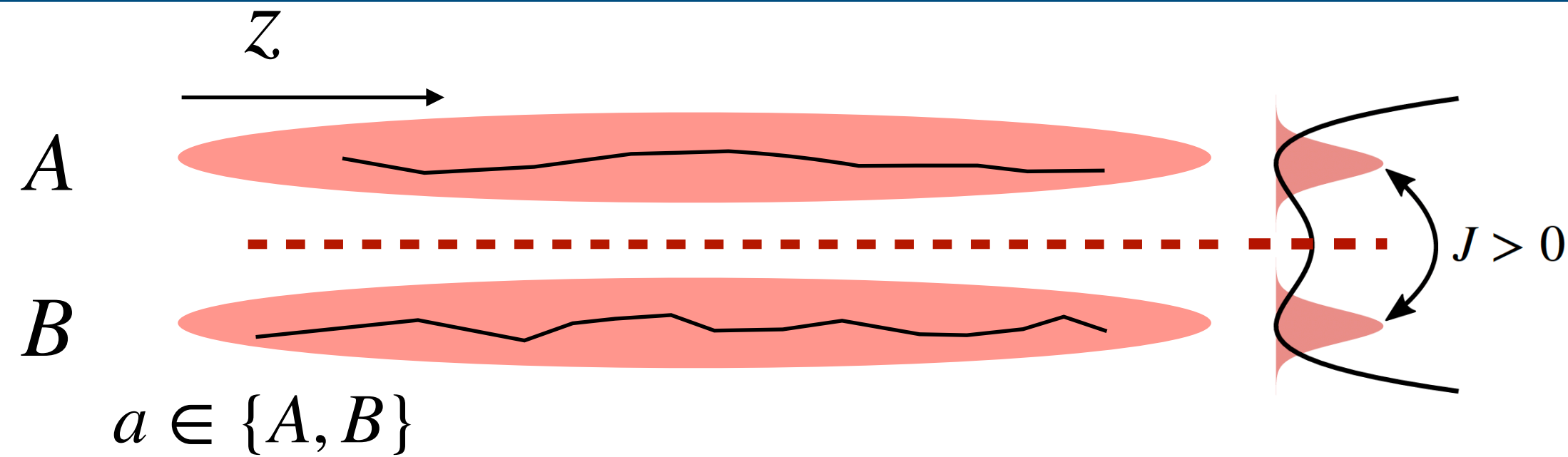


$$\sigma_k \geq 1/2 \quad \forall k$$

Symplectic transform leaves symplectic eigenvalues $\{\sigma_k\}$ invariant

$$\{\sigma_k\} = |\text{eig}[i(\Omega \oplus \Omega)\Gamma]|$$

Mutual Information (MI)



$$I(A : B) = S(A) + S(B) - S(AB)$$

$$S(\Gamma) = \sum_k \left(\sigma_k + \frac{1}{2} \right) \log \left(\sigma_k + \frac{1}{2} \right) - \left(\sigma_k - \frac{1}{2} \right) \log \left(\sigma_k - \frac{1}{2} \right)$$

σ_k : Symplectic eigenvalues of covariance matrix Γ

- $I(A : B)$ **includes both entanglement and classical correlations in general**
- In the limit of **zero temperature** $I(A : B) = 2S(\rho)$ where $S(\rho)$ is the entanglement entropy.

Computing Logarithmic Negativity (LN)

- **Logarithmic negativity (LN)** is directly related to **positive partial transpose (PPT)** criterion for separable states.

- Partial transpose for continuous variable Gaussian states

$$\langle x_1, x_2 | \rho^{\top_B} | x'_1, x'_2 \rangle = \langle x_1, x'_2 | \rho | x'_1, x_2 \rangle$$



$$\tilde{\Gamma}^{\top_B} = \tilde{\Gamma}(\hat{\phi}_k^B \rightarrow -\hat{\phi}_k^B)$$

$$W(q_1, p_1, q_2, p_2) \rightarrow W(q_1, p_1, q_2, -p_2)$$

- Compute symplectic eigenvalues of $\tilde{\Gamma}^{\top_B}$

$$\{\gamma_k\} = \text{eig}[i(\Omega \oplus \Omega)\tilde{\Gamma}^{\top_B}]$$

Logarithmic Negativity (LN) as measure of entanglement

- Compute log. neg. by

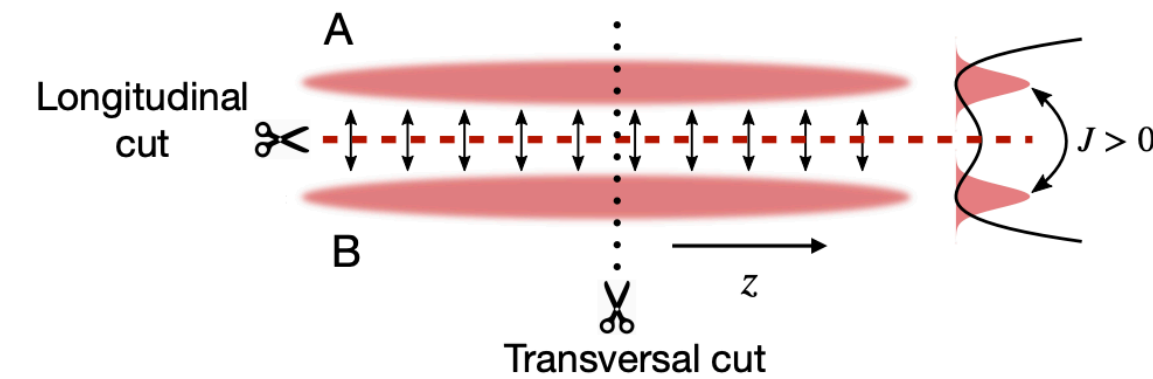
$$E_{\mathcal{N}} = \sum_k \max\{0, -\log(2\gamma_k)\}$$

- $E_{\mathcal{N}} > 0$ is a *sufficient* condition for entanglement. It is *necessary and sufficient* for *two-modes entanglement*
- In our case, we have to solve *multiple independent two-mode problems* (one for each momentum mode k) $\rightarrow E_{\mathcal{N}} > 0$ is necessary and sufficient for entanglement

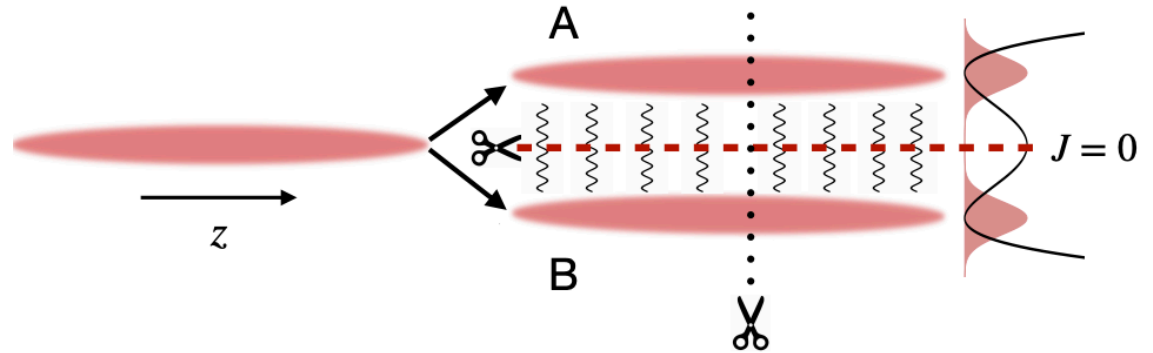
Field-space entanglement in parallel 1D Bose Gases

Murtadho, T., Gluza, M., & Ng, N. H. Y. *Quantum Sci. Technol.* 11: 035060 (2026)

(a) Tunnel coupling



(b) Coherent splitting



* Probe entanglement tunnel-coupling in **Klein-Gordon** (quadratic) at finite T. and **coh. splitting**

* Entanglement in (Gaussian) mixed states → **logarithmic negativity (LN)**

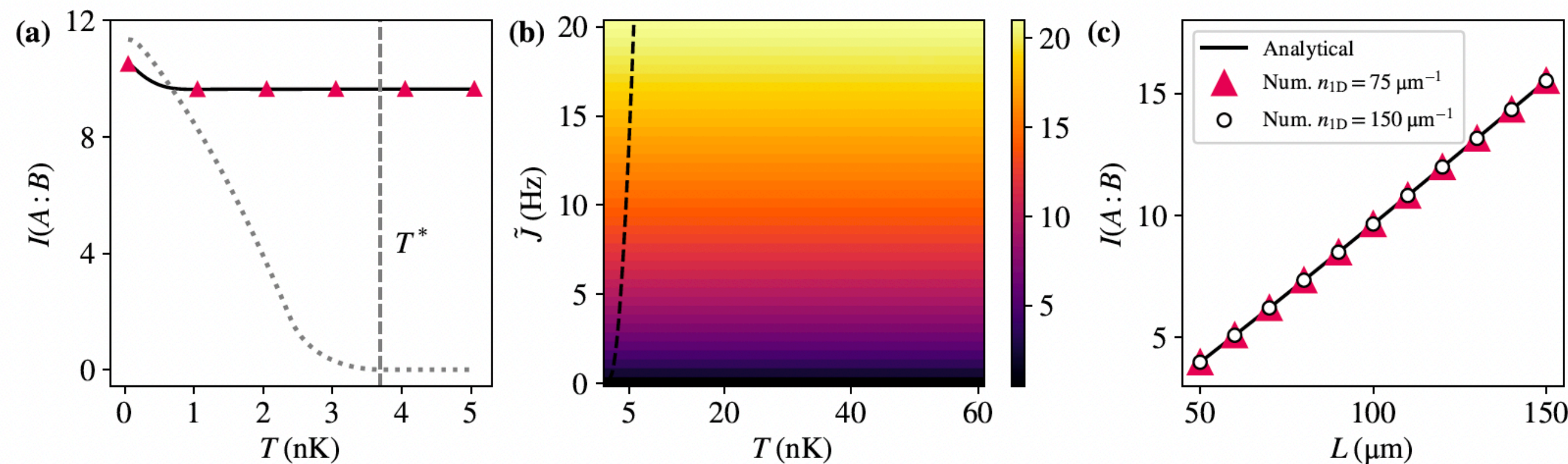
* Total correlation → **Mutual information (MI)**

	Logarithmic Negativity $E_{\mathcal{N}}$	Mutual Information $I(A : B)$
Tunnel coupling	$E_{\mathcal{N}} = \sum_{k>0}^{\Lambda} \max \left\{ 0, -\log \left[\sqrt{C_{k,J}(1 + 2\eta_k^+)(1 + 2\eta_k^-)} \right] \right\}$	$I(A : B) = \sum_{k>0}^{k_{\Lambda}} 2 \left[\left(\lambda_{k,J} + \frac{1}{2} \right) \log \left(\lambda_{k,J} + \frac{1}{2} \right) - \left(\lambda_{k,J} - \frac{1}{2} \right) \log \left(\lambda_{k,J} - \frac{1}{2} \right) \right] \\ - \sum_{a=\pm} \sum_{k>0}^{k_{\Lambda}} [(\eta_k^a + 1) \log (\eta_k^a + 1) - \eta_k^a \log \eta_k^a], \quad (25)$
Coh. Splitting	$E_{\mathcal{N}} = \sum_{a=\pm 1} \sum_{k>0}^{\Lambda} \max \left\{ 0, -\log \left[\sqrt{C_{k,r}^a(1 + 2\eta_k^+)} \right] \right\}$	$I(A : B) = \sum_{k>0}^{k_{\Lambda}} 2 \left[\left(\lambda_{k,r} + \frac{1}{2} \right) \log \left(\lambda_{k,r} + \frac{1}{2} \right) - \left(\lambda_{k,r} - \frac{1}{2} \right) \log \left(\lambda_{k,r} - \frac{1}{2} \right) \right] \\ - (1 + \eta_k^+) \log(1 + \eta_k^+) + \eta_k^+ \log \eta_k^+, \quad (31)$

How “cold” is too “hot” for entanglement?

Murtadho, T., Gluza, M., & Ng, N. H. Y. *Quantum Sci. Technol.* 11: 035060 (2026)

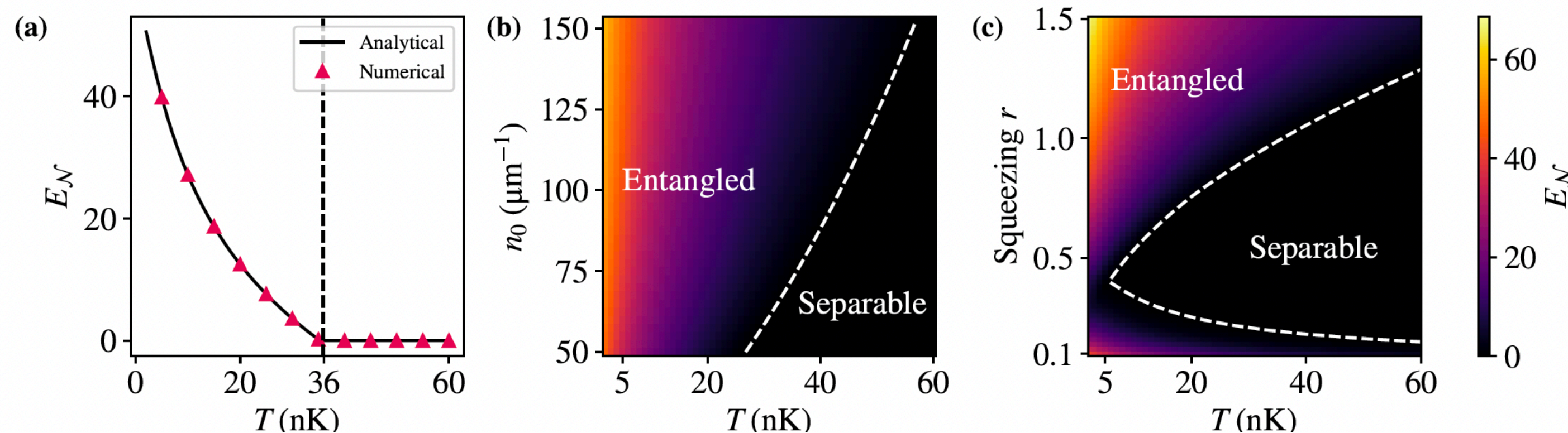
Tunnel-coupled



✱ Entanglement “sudden death”, MI constant $\rightarrow$ Quantum corr. transmutes into classical corr. (emergence of classicality)

✱ Entanglement/MI scales extensively with length, but not with density $\rightarrow$ collective effect

Coherently-split



$$T^* = \sup_{0 < k \leq k_\Lambda} \left\{ T_k^* = \frac{\varepsilon_k^+}{k_B} \left(\ln \left[\sqrt{\frac{C_{k,r} + C_{k,r}^{-1} + 2}{C_{k,r} + C_{k,r}^{-1} - 2}} \right] \right)^{-1} \right\}.$$

Entanglement can be detected at temperatures accessible in the current experiments!

Reconstructing entanglement with (Gaussian) tomography

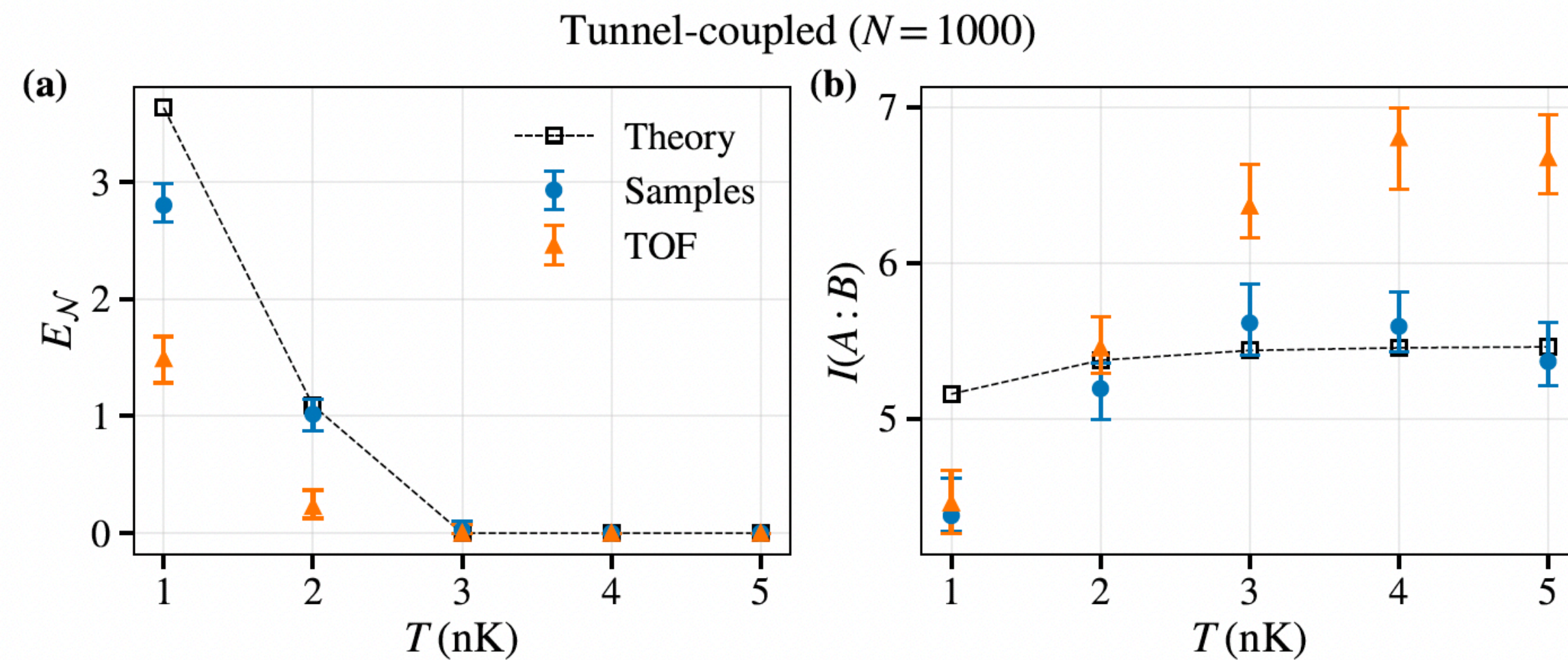


FIGURE 6.7: Tomographic reconstruction of logarithmic negativity (LN) and mutual information (MI) for tunnel-coupled gases in equilibrium. (a) LN vs.

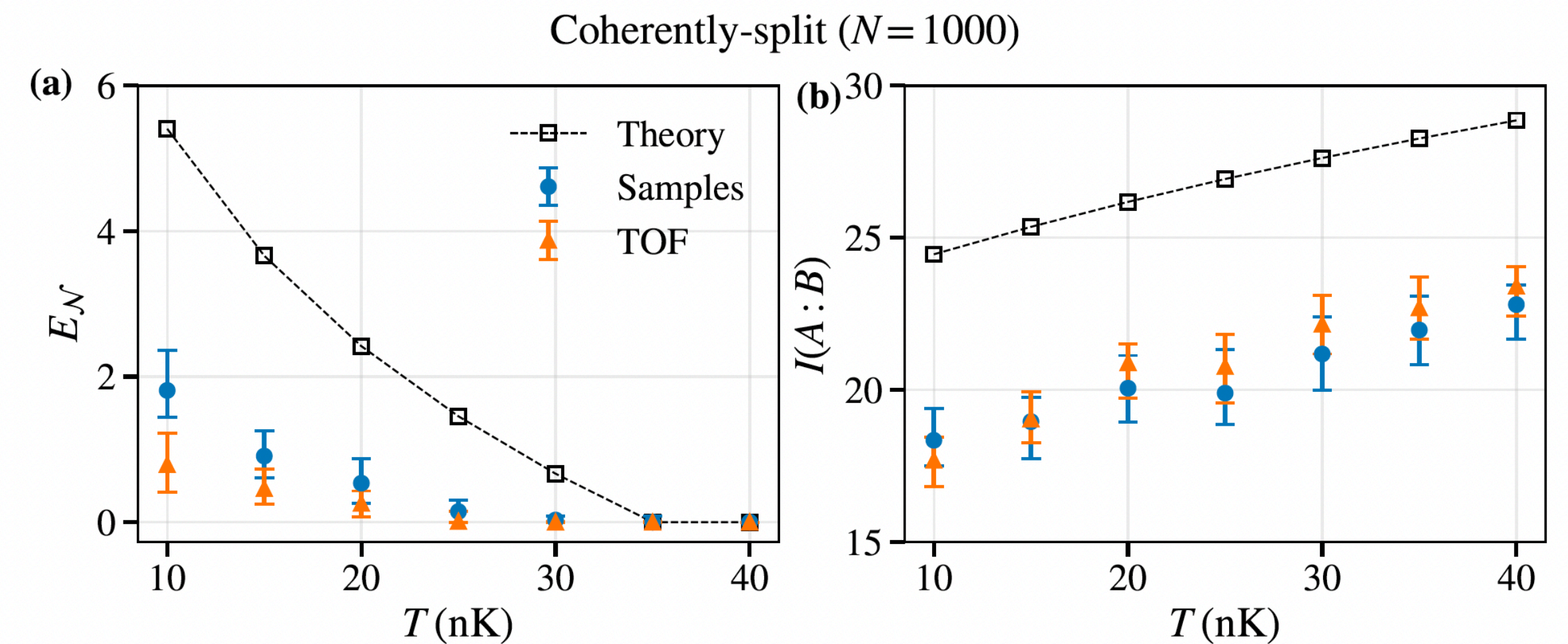


FIGURE 6.8: The same as Fig. 6.7 except that the theoretical covariance matrix is set to be that of after coherent splitting.

$$\{\phi_{\pm}(z, t)\} \rightarrow \Gamma(t) \rightarrow \begin{cases} E_{\mathcal{N}} \\ I(A : B) \end{cases}$$

Outline

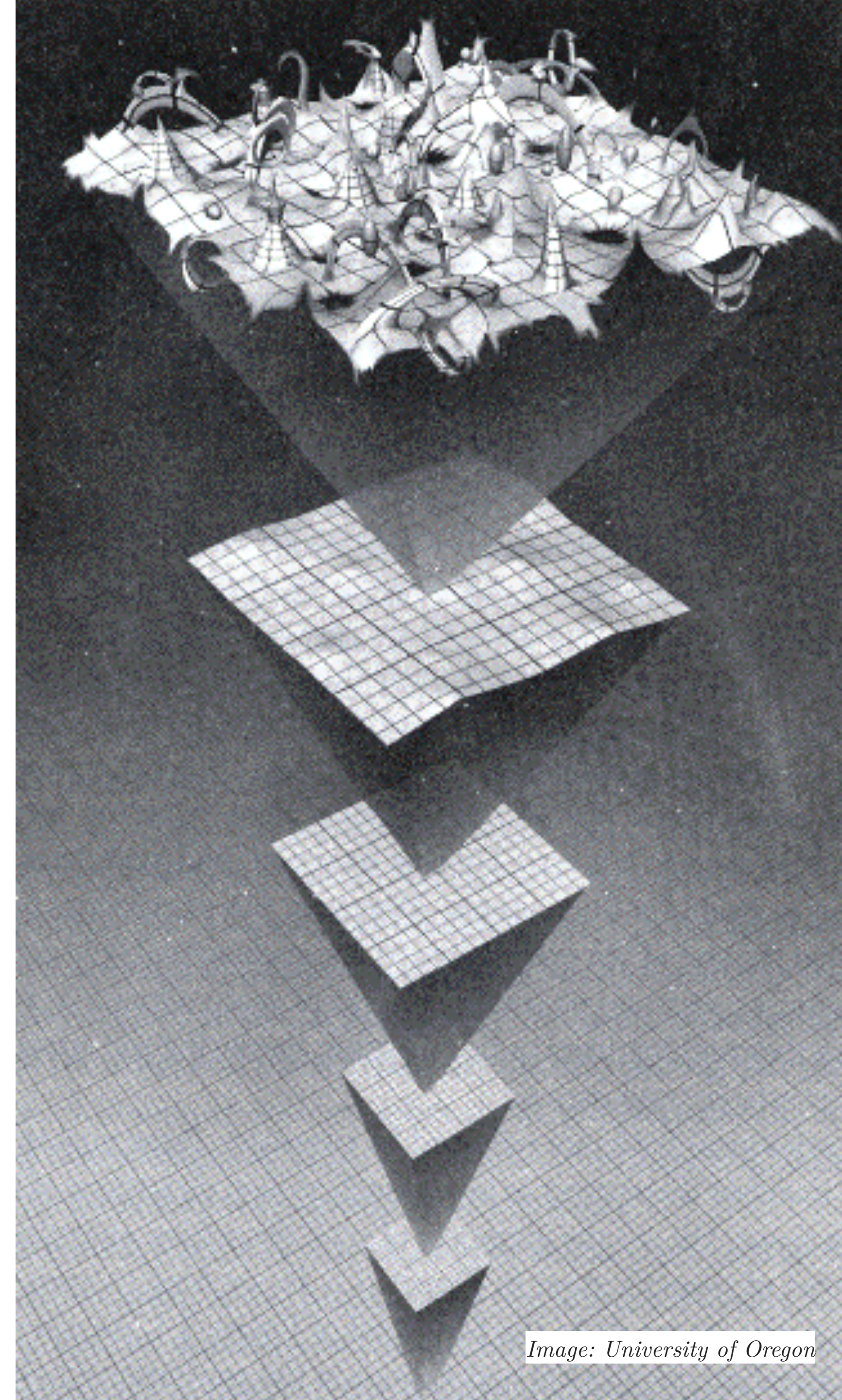
Motivation

Parallel 1D Bose Gases

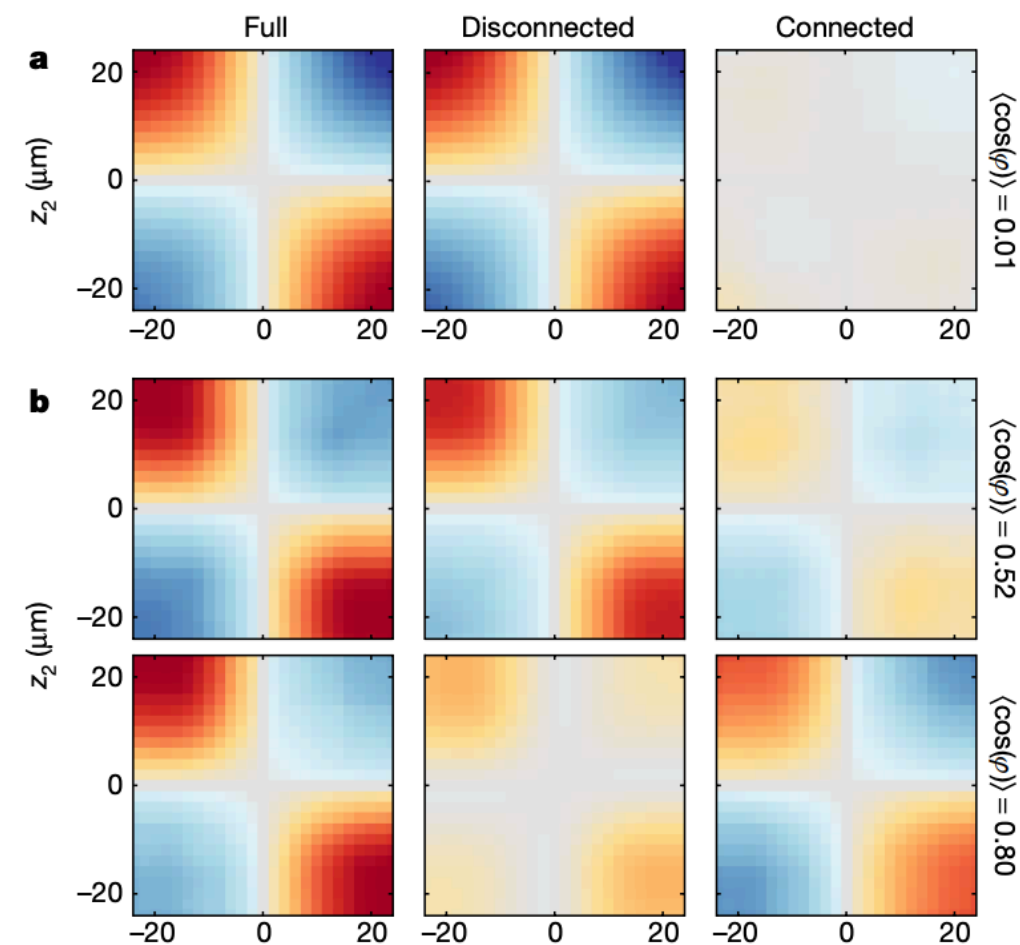
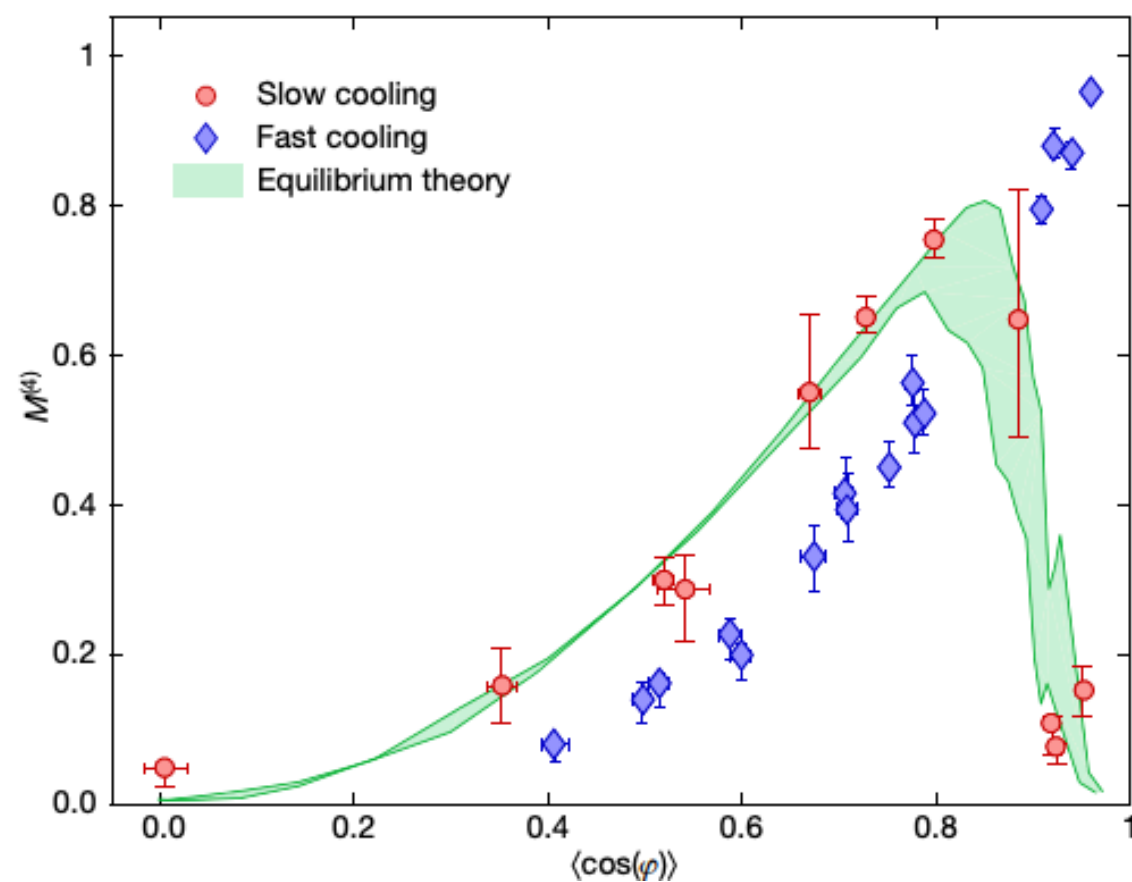
Result: How to Measure Common Phase

Result: Field-space entanglement at finite temperatures

Open Questions and Outlook



Field-space entanglement in the sine-Gordon regime

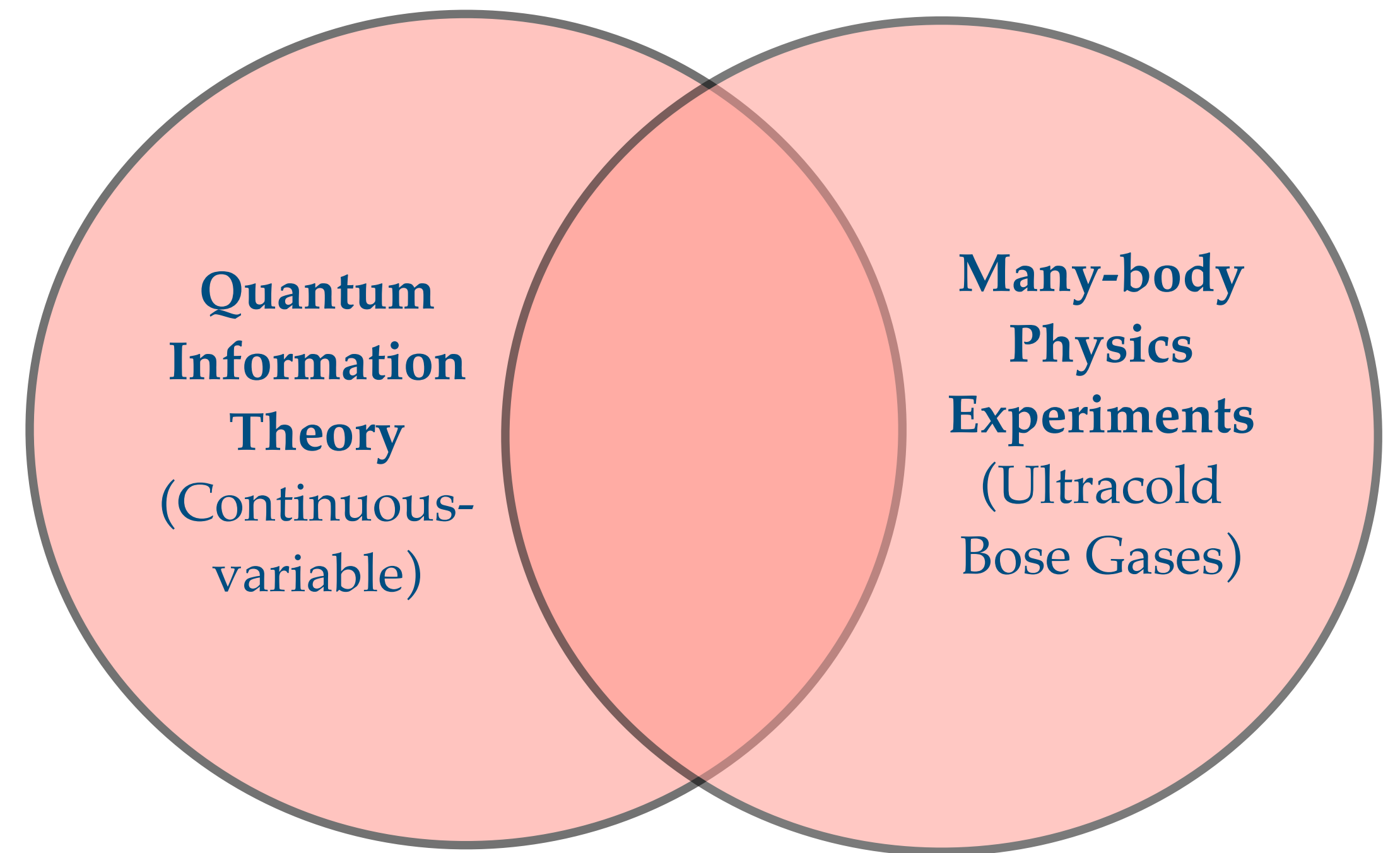


$$G_N(\mathbf{z}; z_0) = \left\langle \prod_{i=1}^N (\hat{\phi}_-(z_i) - \hat{\phi}_-(z_0)) \right\rangle$$

$$G_N(\mathbf{z}; z_0) = G_N^{(\text{disc.})}(\mathbf{z}; z_0) + G_N^{(\text{con.})}(\mathbf{z}; z_0)$$

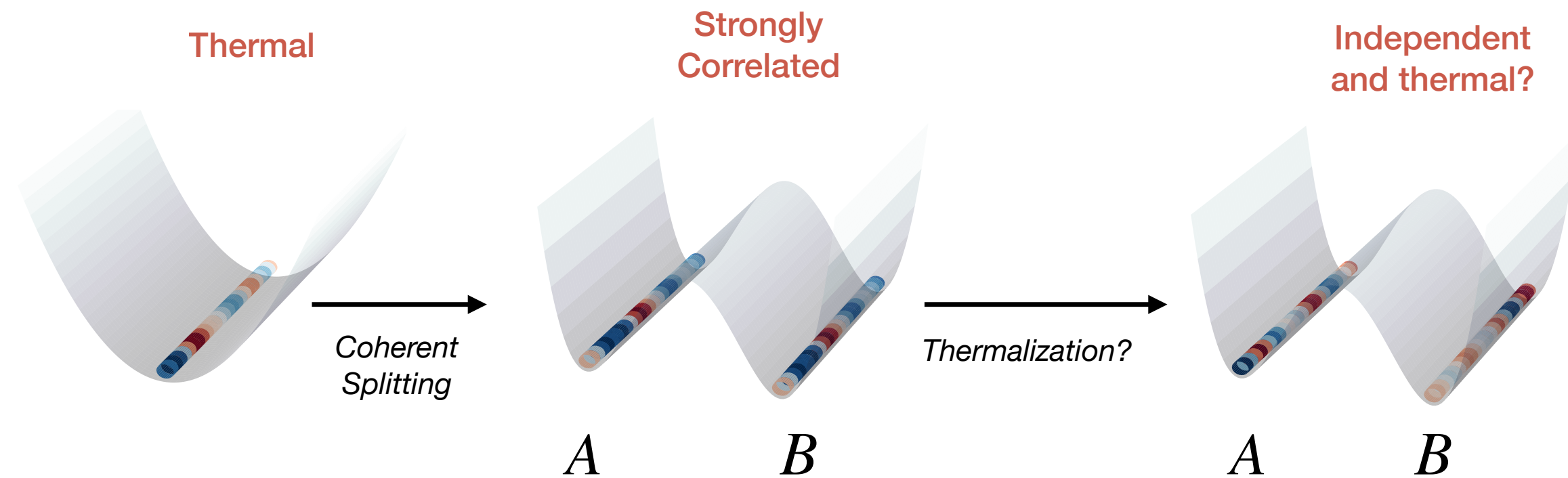
Schweigler, T., et al. *Nature* 545.7654: 323-326 (2017).

Schweigler, T., et al. *Nat. Phys.* 17.5: 559-563 (2021).



Q: Can we measure/infer non-Gaussian CV entanglement from correlation functions?

Where did entanglement escape during thermalization?



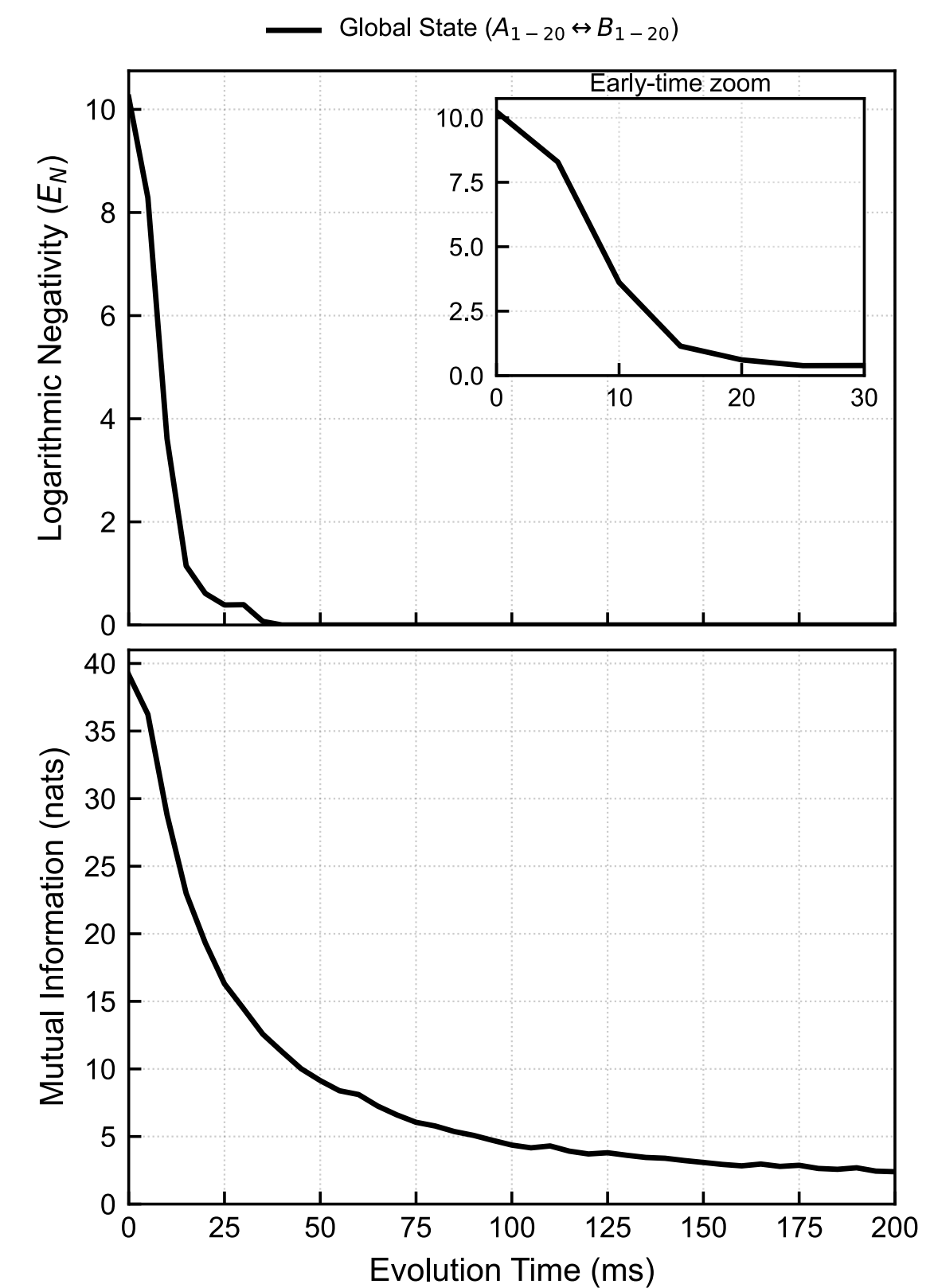
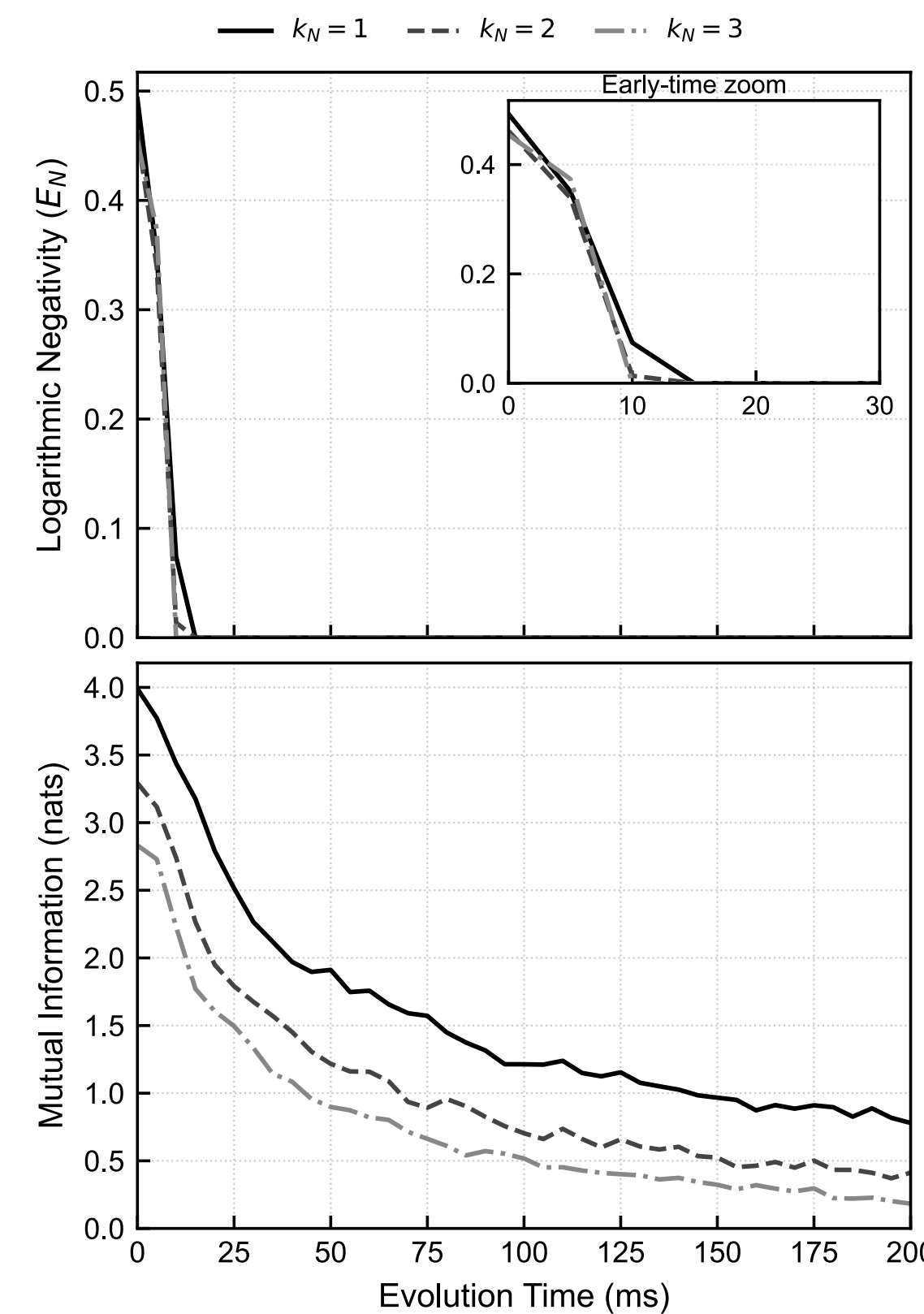
$$U(t) = U_A(t) \otimes U_B(t)$$

Local operators *shouldn't*
change entanglement

**Hypothesis: Leakage of entanglement
from Gaussian to non-Gaussian sector
→ how to quantify this?**

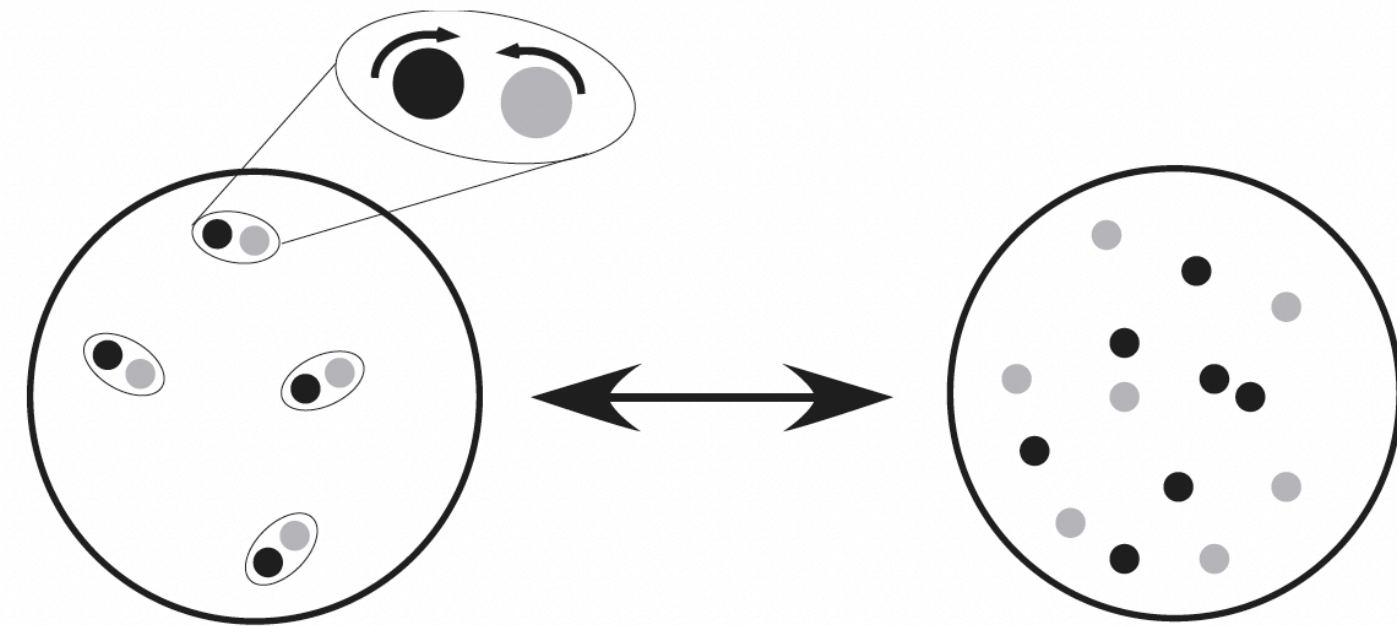
Truncated Wigner Simulation

$$\hat{\Psi}(z, t) \rightarrow \Psi(z, t)$$



Plots Credit: Ishtiak Golam

Extension to higher dimensions (2D)



- * **Superfluid:** vortex is always bound with anti-vortex
- * **Normal fluid:** proliferation of free vortices

} **BKT Transition**

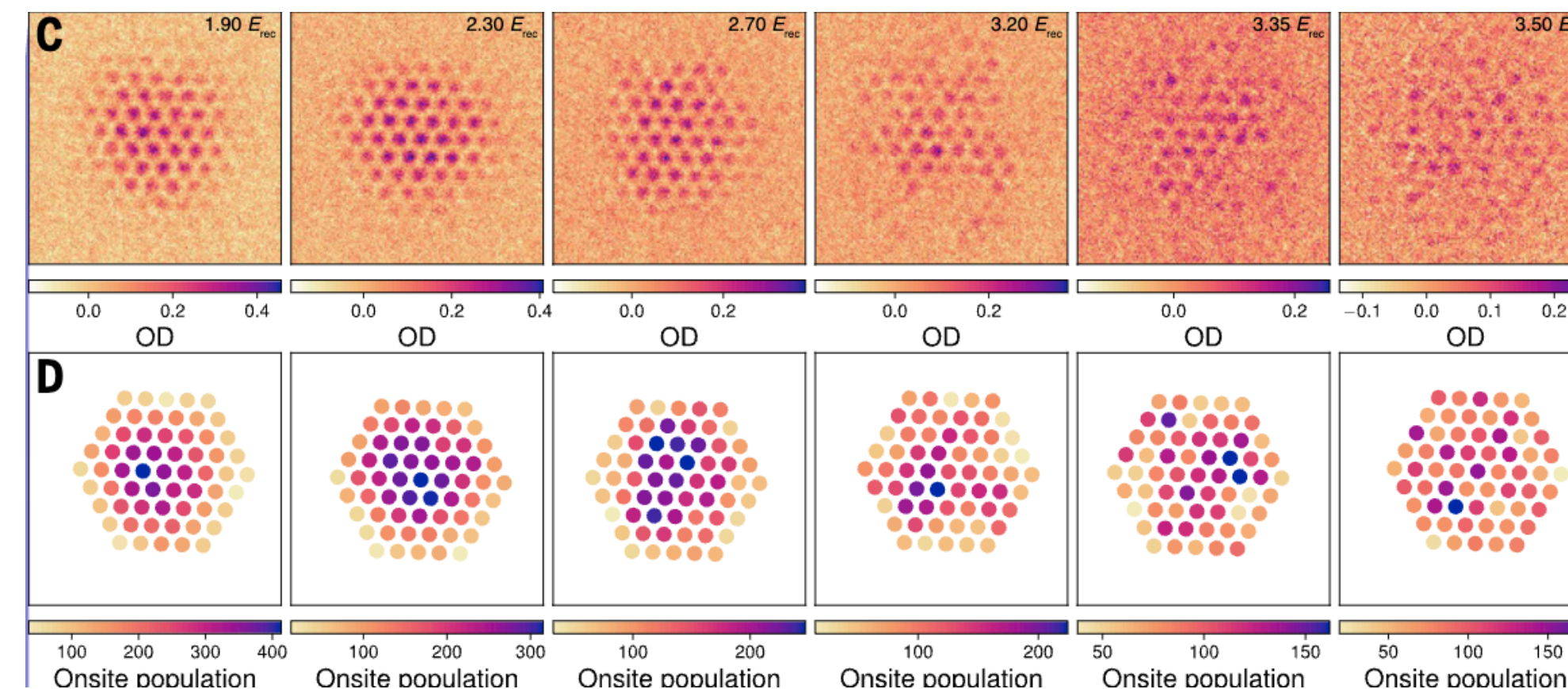
Hadzibabic, Z., & Dalibard, J. *La Rivista del Nuovo Cimento*, 34(6), 389-434 (2011).

RESEARCH ARTICLES

QUANTUM MATTER

A phase microscope for quantum gases

J. C. Brüggenjürgen^{1†}, M. S. Fischer^{1‡}, C. Weitenberg^{1,2*}



Brüggenjürgen, J. C., Fischer, M. S., & Weitenberg, C. *Science*, 393(6807), 167-171 (2026).

Q: Does entanglement play any role in BKT transition?

Take-home messages

Common phase extraction and full Gaussian tomography enables investigation of entanglement in finite-temperature quantum field simulators

State-of-the-art experiment can already probe entanglement with coherent splitting

We live in exciting era for quantum field simulation!

Theory: Marek Gluza, Nelly Ng

Experiment: Jörg Schmiedmayer, Sebastian Erne,
Federica Cataldini, Mohammadamin Tajik



Image: University of Oregon

